

Physics — Final Exam

Practice Questions with Answers and Explanations

Question: 188

Language: English

How to use this book

This collection contains all 188 questions of the test “Physics — Final Exam”. Every question shows all answer choices, the correct answer highlighted in green, and a detailed explanation below it. Try to answer on your own first, then check yourself. The answer key on the last page allows quick self-testing.

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1. A ballistic pendulum consists of a large block of wood of mass $M = 9.99$ kg. A bullet of mass $m = 10$ g, travelling horizontally with speed v , hits the pendulum and remains stuck in it. After the collision, the pendulum oscillates and reaches a maximum height $h = 5$ cm. What is the velocity v of the bullet?

- A. $v = 5 \times 10^4$ m/s
- B. $v = 102$ m/s
- C. $v = 103$ m/s ✓
- D. $v = 104$ m/s

EXPLANATION

Conservation of momentum and energy: $m \cdot v = (M+m) \cdot V$, then $(M+m) \cdot g \cdot h = 0.5 \cdot (M+m) \cdot V^2$.

2. A ballistic pendulum is a device used to calculate the muzzle velocities of bullets. It is a pendulum constructed by hanging a wooden block with a very high mass at the end of a rope with length L . When a bullet with mass m traveling at speed v embeds itself into the wooden block with mass M , the (bullet + block) system rises by height h . Calculate the speed of the bullet when: $m = 50$ g, $M = 5$ kg, $L = 2$ m, $h = 80$ cm

- A. $v = 4000$ m/s
- B. $v = 4$ m/s
- C. $v = 40$ m/s
- D. $v = 400$ m/s ✓

EXPLANATION

Conservation of momentum and energy: $m \cdot v = (M+m) \cdot V$, then $(M+m) \cdot g \cdot h = 1/2 \cdot (M+m) \cdot V^2$.

3. A block of mass $m = 1\text{kg}$ is attached to one end of a rope going around a pulley. Another 1 block of mass $m = 2\text{kg}$ at rest on a frictionless horizontal plane is tied to the other end of the 2 rope. The pulley has mass $M = 2\text{kg}$, radius $R = 20\text{cm}$, and moment of inertia $I = 3\text{kg} \cdot \text{m}^2$. Calculate the linear acceleration a and angular acceleration α .

- A. $a = 0.33 \text{ m}; \alpha = 1.64\text{s}^{-2} \text{ s}^2$
- B. $a = 0.53 \text{ m}; \alpha = 2.64\text{s}^{-2} \text{ s}^2$ ✓
- C. $a = 0.13, \alpha = 0.64\text{s}^{-2} \text{ s}^2$
- D. $a = 1.13 \text{ m}; \alpha = 0.64\text{s}^{-2} \text{ s}^2$

EXPLANATION

$a = (m_2g) / (m_1 + m_2 + I/R^2) \approx 0.53 \text{ m/s}^2; \alpha = a/R \approx 2.64 \text{ rad/s}^2$. B) $a = 0.53 \text{ m}; \alpha = 2.64\text{s}^{-2} \text{ s}^2$

4. A block of mass $m = 1\text{kg}$ is attached to one end of a rope going around a pulley. Another block of mass $m = 2\text{kg}$, at rest on a frictionless horizontal plane, is tied to the other end of the rope. The pulley has a mass $M = 2\text{kg}$, a radius $R = 20\text{cm}$, and a moment of inertia $I = 3\text{kg} \cdot \text{m}^2$. Calculate the tensions T_1 and T_2 in the ropes.

- A. $T_1 = 0.9\text{N}; T_2 = 0.3\text{N}$
- B. $T_1 = 9.9\text{N}; T_2 = 9.3\text{N}$
- C. $T_1 = 9.9\text{N}; T_2 = 3.3\text{N}$
- D. $T_1 = 9.9\text{N}; T_2 = 0.3\text{N}$ ✓

EXPLANATION

Key equation: $(T_2 - T_1) * R = I * \alpha$.

5. In a ballistic pendulum experiment, a bullet of mass $m = 50\text{g}$ is fired with $v = 400 \text{ m/s}$ into a stationary wooden block of mass $M = 5\text{kg}$ hanging from a string of length $L = 2\text{m}$. The bullet embeds itself in the block, and the combined system swings upward to a maximum height above the initial position. What is the maximum angle θ (from the vertical) to which the pendulum rises?

- A. $\theta = 30.0^\circ$
- B. $\theta = 66.4^\circ$
- C. $\theta = 45.6^\circ$ ✓
- D. $\theta = 53.1^\circ$

EXPLANATION

Conservation of energy: $0.5*m*v^2 = (M+m)*g*L*(1 - \cos\theta)$.

6. A block of 0.1kg of ice at $T = 0^\circ\text{C}$ is completely melted always at temperature $T = 0^\circ\text{C}$. What is its variation of entropy ΔS ? (The latent heat of fusion of water is $L = 3.335 \times 10^5 \text{J/kg}$)

- A. $\Delta S = 0 \text{ J}$ because this thermal transformation is reversible
- B. $\Delta S = 1900$
- C. $\Delta S = -190 \text{ J}$
- D. $\Delta S = 190 \text{ J}$ ✓

EXPLANATION

Entropy change: $\Delta S = m*L/T$.

7. A block of 0.1kg of ice at $T = 0^\circ\text{C}$ is completely melted always at temperature $T = 0^\circ\text{C}$. What is its variation of entropy ΔS ? (The latent heat of fusion of water is $L = 3.335 \times 10^5 \text{ J/kg}$)

- A. $\Delta S = 0 \text{ J}$ because this thermal transformation is reversible
- B. $\Delta S = 122 \text{ J}$ ✓
- C. $\Delta S = 1220$
- D. $\Delta S = -122 \text{ J}$

EXPLANATION

Entropy change: $\Delta S = m \cdot L / T$.

8. A body of mass m and specific heat c , initially at temperature T_i , exchanges heat from a single hot reservoir (at constant temperature T_h). The body is not a reservoir, so it has a finite heat capacity, and its temperature will increase up to T_h . What is the entropy variation of the Universe (system + environment)?

- A. $\Delta S = 0$
- B. $\Delta S = mc(T_h - T_i) + mc \ln(T_h)$
- C. $\Delta S = mc(T_i - T_h) + mc \ln(T_i)$
- D. $\Delta S = mc(T_i - T_h) + mc \ln(T_h)$ ✓

EXPLANATION

Entropy change: $\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$.

9. A body of mass $M = 10\text{kg}$ falls from a height $h = 10\text{m}$ and stops. Initially the body and the environment are at the same temperature $T = 293\text{K}$. What is the entropy variation ΔS of the universe?

- A. $\Delta S = 33.3\text{J/K}$
- B. $\Delta S = -3.3\text{J/K}$
- C. $\Delta S = 0\text{J/K}$
- D. $\Delta S = 3.3\text{J/K}$ ✓

EXPLANATION

Entropy change: $\Delta S = Q/T = m \cdot g \cdot h / T$.

10. A bullet with mass $m = 50\text{g}$ travels at a speed of $v = 200\text{m/s}$ and embeds itself into the rim of a disk at rest that can rotate freely, at a distance of $R = 60\text{cm}$ from the center. The disk mass is $M = 0.9\text{kg}$. What is the angular speed ω of the system ($M + m$) after the collision?

- A. $\omega = 33\text{s}^{-1}$ ✓
- B. $\omega = 3.3\text{s}^{-1}$
- C. $\omega = 0.3\text{s}^{-1}$
- D. $\omega = 6.0\text{s}^{-1}$

EXPLANATION

Conservation of angular momentum: $m \cdot v \cdot R = (M \cdot R^2 + m \cdot R^2) \cdot \omega$.

11. A Carnot engine, operating between reservoirs at 20°C and 200°C , produces 10kW of power. The rejected heat per unit time is nearest to:

- A. 26.3 kJ/s
- B. 20.2 kJ/s
- C. 16.3 kJ/s ✓
- D. 12.0 kJ/s

EXPLANATION

Carnot efficiency: $\eta = 1 - T_{\text{Cold}}/T_{\text{hot}}$, $Q_{\text{out}} = Q_{\text{in}} \cdot (1 - \eta)$.

12. A Carnot refrigerator requires 10kW to remove 20kJ/s from a 20°C reservoir. The temperature of the high-temperature reservoir is nearest:

- A. 440K ✓
- B. 400K
- C. 360K
- D. 320K

EXPLANATION

$\text{COP} = Q_{\text{c}}/(W) = T_{\text{c}}/(T_{\text{h}} - T_{\text{c}})$, solve for T_{h} .

13. A conservative system goes from state A to state B and the change in the potential energy is equal to 200J. Which of the following statements is correct?

- A. The variation of kinetic energy is 400J
- B. The variation of kinetic energy is 200J
- C. The work of the forces is equal to -200J ✓
- D. The work of the forces is equal to 200J

EXPLANATION

Work-energy theorem: $W = -\Delta U = -200\text{J}$.

14. A copper block of mass $M = 0.5\text{kg}$ falls from a height $h = 100\text{m}$ into a lake at temperature $T = 283\text{K}$. The initial temperature of the copper block is $T = 423\text{K}$. Calculate the change ΔS in entropy of the universe ΔS in this process. The specific heat of copper is $c = 387\text{J/kg} \cdot \text{K}$. U

- A. $\Delta S = 9.71 \text{ J}$ ✓
- B. $\Delta S = 140.00 \text{ J}$
- C. $\Delta S = -9.71 \text{ J}$
- D. $\Delta S = 19.7 \text{ J}$

EXPLANATION

$\Delta S_{\text{universe}} = \Delta S_{\text{copper}} + \Delta S_{\text{water}} = Mc\Delta T/T + m_{\text{water}}c_{\text{water}}\Delta T/T \approx 9.71 \text{ J}$. A) $\Delta S = 9.71 \text{ J}$

15. A gas expands from volume $V = 1\text{m}^3$ to volume $V = 3\text{m}^3$ along the parabolic curve $P = 3V^2 - 1/2$ (in Pascals), as shown in the figure. Calculate the work W performed by the gas.

- A. $W = 0.26\text{J}$
- B. $W = 260\text{J}$
- C. $W = 2.6\text{J}$
- D. $W = 26\text{J}$ ✓

EXPLANATION

Work = $\int P dV = \int 3V^2 dV$ from 1 to 3.

16. A gas is compressed at constant temperature. Which is correct?

- A. It will receive heat.
- B. It will give off heat. ✓
- C. Its internal energy will increase.
- D. Its internal energy will decrease.

EXPLANATION

First law: $\Delta U = Q - W$, for isothermal process $\Delta U = 0$, so $Q = W$.

17. A gas with $\gamma = c_p(P) = 1.4$ undergoes a 3-step cycle, as shown in the figure. $c_p(V)$ It expands adiabatically along the path ab, is compressed under constant pressure along the path bc, and is heated at constant volume along the path ca. Temperatures: $T_a = 500\text{K}$, $T_b = 400\text{K}$, $T_c = 300\text{K}$ Calculate the efficiency η of the cycle.

- A. $\eta = 15\%$
- B. $\eta = 0.3\%$
- C. $\eta = 3\%$
- D. $\eta = 30\%$ ✓

EXPLANATION

Efficiency: $\eta = 1 - Q_{\text{out}}/Q_{\text{in}} = 1 - (T_c/T_h) * (V_c/V_h)^{\gamma-1}$.

18. A helium-filled rubber balloon is left in a car on a cold winter night. Compared to its size when it was in the warm car the afternoon before, the size the next morning is:

- A. larger
- B. smaller ✓
- C. unchanged
- D. Not enough information to say

EXPLANATION

Charles' law: $V \propto T$, temperature decreases, volume decreases.

19. A homogeneous solid cylinder of length $l = 40\text{cm}$, radius $r = 2\text{cm}$, and mass $m = 2\text{kg}$ rotates in a horizontal plane around a vertical axis perpendicular to its axis through its midpoint O, as a result of an impulse $J = 5 \cdot 10^{-2}\text{N} \cdot \text{s}$ received perpendicularly at the edge. As a result of friction, the cylinder stops after traveling one lap. The initial angular speed ω is 0.37s^{-1} . Calculate the moment M of the friction forces.

- A. $M = 2.93 \cdot 10^{-3}\text{Nm}$
- B. $M = 2.93 \cdot 10^{-4}\text{Nm}$
- C. $M = 2.93\text{Nm}$
- D. $M = 2.93 \cdot 10^{-2}\text{Nm}$ ✓

EXPLANATION

Angular deceleration: $\alpha = \Delta \omega / \Delta t$, moment $M = I \cdot \alpha$.

20. A homogeneous solid cylinder of length $l = 40\text{cm}$, radius $r = 2\text{cm}$, and mass $m = 2\text{kg}$ rotates in a horizontal plane around a vertical axis perpendicular to its axis through the middle point O, as a result of an impulse $J = 5 \cdot 10^{-2}\text{N} \cdot \text{s}$ received perpendicularly at an extreme (the rim). As a result of the friction, the cylinder stops after traveling a lap. Calculate the initial angular speed ω of the cylinder.

- A. $\omega = 3.70\text{s}^{-1}$
- B. $\omega = 0.07\text{s}^{-1}$
- C. $\omega = 0.37\text{s}^{-1}$ ✓
- D. $\omega = 37.00\text{s}^{-1}$

EXPLANATION

Angular momentum: $J = I \cdot \omega_0$.

21. A homogeneous spherical mass M of radius R , when $r < R$, produces a gravitational field $G(r)$ equal to: (here r is the distance from the center of the sphere) γM^*

- A. $G(r) = -GM/R^3r$
- B. $G(r) = -GM/R^2r$ ✓
- C. $G(r) = -GM/r^2r$
- D. $G(r) = -GM/r^3r$

EXPLANATION

Inside a sphere, gravitational field: $G(r) = (GM/R^3)r$.

22. A homogeneous superconducting cylinder is suspended in air in vertical position relative to ground, with mass M and radius R , and rotates around its barycenter axis with initial angular velocity ω and initial angular position θ_0 . It is subjected to a braking moment $M = -k\omega$, where k is a constant and ω is the instantaneous angular velocity. Derive the laws of motion $\theta(t)$ and $\omega(t)$ as functions of the assigned variables.

- A. $\theta(t) = \theta_0 + MR^2\omega/(1-\exp(-2kt/MR^2))$; $\omega(t) = \omega_0 \exp(-2kt/MR^2)$
- B. $\theta(t) = \theta_0 + \omega_0 \exp(-2kt/MR^2)$; $\omega(t) = \omega_0 \exp(-2kt/MR^2)$
- C. $\theta(t) = \theta_0 + MR^2\omega_0/(1-\exp(-2kt/MR^2))$; $\omega(t) = \omega_0 \exp(-2kt/MR^2)$
- D. $\theta(t) = \theta_0 + (1-\exp(-2kt/MR^2))/2k$; $\omega(t) = \omega_0 \exp(-2kt/MR^2)$ ✓

EXPLANATION

Angular motion: $\theta(t) = \theta_0 + \omega_0 \cdot t - (k/(2 \cdot I)) \cdot t^2$.

23. A mass $m = 1\text{kg}$ is attached to the end of a rope that is wrapped around a pulley with radius $R = 0.2\text{m}$ and moment of inertia $I = 0.5\text{kg} \cdot \text{m}^2$. The mass is released from a height $h = 2.7\text{m}$. With what velocity v will it hit the ground?

- A. $v = 2\text{m/s}$ ✓
- B. $v = 0.2\text{m/s}$
- C. $v = 20\text{m/s}$
- D. $v = 12\text{m/s}$

EXPLANATION

Energy conservation: $m \cdot g \cdot h = 0.5 \cdot m \cdot v^2 + 0.5 \cdot I \cdot (v/R)^2$.

24. A mass M produces, in the external space, a gravitational field $M G(r) = -\gamma/r^2 \hat{u}_r$ where r is the distance from the center of mass. Under which condition is this field expression valid?

- A. Always
- B. Just if it is a point mass or a homogeneous spherical mass ✓
- C. Just if a solid mass
- D. Just if a macroscopic mass

EXPLANATION

The field expression holds only for point masses or homogeneous spherical masses (shell theorem). B) Just if it is a point mass or a homogeneous spherical mass

25. A mole of an ideal monoatomic gas travels along the cycle shown in the figure. Calculate the variations in internal energy ΔU and entropy ΔS over a complete cycle.

- A. $\Delta U = 3pV \text{ J}; \Delta S = -\text{J/K}$
- B. $\Delta U = 3pV \text{ J}; \Delta S = \text{J/K}$
- C. $\Delta U = -3pV \text{ J}; \Delta S = 0 \text{ J/K}$
- D. $\Delta U = 0 \text{ J}; \Delta S = 0 \text{ J/K}$ ✓

EXPLANATION

First law: $\Delta U = Q - W$, for a cycle $\Delta U = 0$.

26. A mole of an ideal gas is initially in the state P, V, T . The gas undergoes the following three-step process: heated at constant volume to a temperature $2T$, expanded isothermally to volume $2V$ and cooled at constant pressure back to temperature T .

What is the total change in entropy ΔS of the gas? $3 \cdot 8.31$

- A. $\Delta S = \ln 2 \text{ J/K}$
- B. $\Delta S = \ln 2 \text{ J/K}$
- C. $\Delta S = \text{J/K}$
- D. $\Delta S = 0 \text{ J/K}$ ✓

EXPLANATION

Entropy change: $\Delta S = nR \ln(V_f/V_i) + nC_v \ln(T_f/T_i)$.

27. A 1.0 mole sample of an ideal gas is kept at 0.0°C during an expansion from 3.0L to 10.0L. • How much work W is done by the gas during the expansion? • How much energy Q is transferred as heat with the surroundings during the process?

- A. $W = 5.4 \times 10^3 \text{ J}; Q = 2.7 \times 10^3 \text{ J}$
- B. $W = 2.7 \times 10^3 \text{ J}; Q = -2.7 \times 10^3 \text{ J}$
- C. $W = 2.7 \times 10^3 \text{ J}; Q = 0 \text{ J}$
- D. $W = 2.7 \times 10^3 \text{ J}; Q = 2.7 \times 10^3 \text{ J}$ ✓

EXPLANATION

Work: $W = nRT \ln(V_f/V_i)$, $Q = W$ for isothermal process.

28. A motorcycle showman with mass m drives through a vertical circular track with radius $r = 40\text{m}$. What should his minimum speed v be at the top point so that the motorcycle does not lose contact with the track?

- A. $v = 20 \text{ m/s}$ ✓
- B. $v = 40 \text{ m/s}$
- C. $v = 10 \text{ m/s}$
- D. $v = 30 \text{ m/s}$

EXPLANATION

Centripetal force: $(mv^2)/r = mg$.

29. A pencil of length $L = 15 \text{ cm}$ and mass m is placed vertically on a rough plane and initially held at rest. It falls by rotating about the point of contact with the ground (acting as a pivot). What is the angular speed ω of the pencil at the instant it hits the ground?

- A. $\omega = 1.40 \text{ s}^{-1}$
- B. I cannot answer because I don't know the pencil mass
- C. $\omega = 0.14 \text{ s}^{-1}$
- D. $\omega = 14 \text{ s}^{-1}$ ✓

EXPLANATION

Energy conservation: $0.5 \cdot I \cdot \omega^2 = m \cdot g \cdot (L/2)$.

30. A perfect gas thermal machine describes a cycle composed of: a reversible isothermal expansion, a reversible isochoric transformation, an irreversible isothermal compression, and a final reversible isochoric transformation. What can be said about the entropy variation in this cycle?

- A. The entropy variation of the thermodynamic universe is zero
- B. The entropy variation of the thermodynamic universe is negative ✓
- C. The entropy variation of the gas is positive
- D. The entropy variation of the gas is zero

EXPLANATION

Irreversible compression increases universe entropy ($\Delta S_{\text{universe}} > 0$). B) The entropy variation of the thermodynamic universe is negative

31. A perfect gas undergoes an isobaric transformation. Let ΔT be the temperature difference between the final and initial states, and let n be the number of moles of gas. What is the variation of internal energy ΔU ?

- A. $\Delta U = C \Delta T$
- B. $\Delta U = nC \Delta T$ ✓
- C. $\Delta U = nC \Delta T$
- D. $\Delta U = C \Delta T$

EXPLANATION

Internal energy change: $\Delta U = n \cdot C_v \cdot \Delta T$.

32. A physical pendulum is constructed by hanging a uniform rod of length $L = 60 \text{ cm}$ and mass m from one of its ends. What is the period T of small oscillations? At what distance d from its center of mass should it be suspended so that the period becomes $T = 2.0 \text{ s}$?

- A. $T = 0.26 \text{ s}$; $d = 0.30 \text{ m}$
- B. $T = 0.26 \text{ s}$; $d = 0.30 \text{ m}$
- C. $T = 1.26 \text{ s}$; $d = 0.03 \text{ m}$ ✓
- D. $T = 1.26 \text{ s}$; $d = 0.30 \text{ m}$

EXPLANATION

Period: $T = 2\pi\sqrt{I/(m \cdot g \cdot d)}$, moment of inertia $I = (1/3) \cdot m \cdot L^2$.

33. A physical pendulum swings around an axis passing through a point P located a distance d from the center of mass. The pendulum has: Mass M , moment of inertia I about the rotation axis, distance d from the center of mass to axis P. What is the period T of small oscillations? (cid:114) I

- A. $T = 2\pi$ ✓
- B. $T = 2\pi$
- C. $T = 2\pi$
- D. $T = 2\pi \sqrt{I/Mgd}$

EXPLANATION

Period: $T = 2\pi \sqrt{I/(m g d)}$.

34. A piece of ice is floating in a glass of water. What will happen to the level of water when the ice fully melts?

- A. It will increase.
- B. It will decrease.
- C. It will remain the same. ✓
- D. It is impossible to tell.

EXPLANATION

Archimedes' principle: floating ice displaces its own weight in water.

35. A piston allows air to expand from $6 \cdot 10^6 \text{Pa}$ to $2 \cdot 10^5 \text{Pa}$. The initial volume and temperature are 500cm^3 and 800°C . Assuming temperature is held constant, calculate: the heat transfer Q and the entropy change ΔS . Assume the molecular weight of air is 28.96g/mol .

- A. $Q = 1.02 \cdot 10^4 \text{J}$; $\Delta S = 9.51 \text{J/K}$ ✓
- B. $Q = 1.02 \cdot 10^3 \text{J}$; $\Delta S = 9.51 \text{J/K}$
- C. $Q = 1.02 \cdot 10^4 \text{J}$; $\Delta S = 95.1 \text{J/K}$
- D. $Q = 1.02 \cdot 10^5 \text{J}$; $\Delta S = 95.1 \text{J/K}$

EXPLANATION

Work: $W = nRT \ln(P_i/P_f)$, entropy: $\Delta S = nR \ln(V_f/V_i)$.

36. A planet describes an elliptical orbit around a star. Let $\omega = d\theta/dt$ be its angular velocity and r be the instantaneous distance from the star. Which of the following relationships is correct?

- A. $\omega \cdot r = \text{constant}$
- B. $\omega/r = \text{constant}$
- C. $\omega/r^2 = \text{constant}$
- D. $\omega \cdot r^2 = \text{constant}$ ✓

EXPLANATION

Kepler's second law: $dA/dt = 0.5 \cdot r^2 \cdot \omega = \text{constant}$.

37. A plastic cube floats in a basin full of water. It is observed that the cube protrudes above the water surface by 20% of its edge length. What is the ratio between the density of the plastic and that of the water ρ_p ? ρ_w

- A. $\rho_p = 0.6$
- B. $\rho_p = 0.4$
- C. $\rho_p = 0.2$
- D. $\rho_p = 0.8$ ✓

EXPLANATION

Buoyancy: $\rho_p \cdot V = \rho_w \cdot 0.8 \cdot V$.

38. A point-like particle, moving at speed v , is subject to a force $\vec{F} = b\vec{x} \times \vec{v}$ where b is a constant vector. What can be said about the work done by this force?

- A. It is zero just when the point particle moves with circular motion.
- B. It is always different from zero.
- C. It is always zero. ✓
- D. It depends on the value of b .

EXPLANATION

Work by magnetic force: $\vec{F} \perp \vec{v} \Rightarrow W = 0$.

39. A point mass $M = 2\text{kg}$ is initially at rest at the origin of a Cartesian coordinate system. At time $t = 0$, a force field $\vec{F}(x,y,z) = k(2x\hat{i} - 4y\hat{j} + 6z\hat{k})$ with $k = 2\text{N/m}$ acts on it, where $\vec{F}$ is in newtons and x,y,z are in meters. Neglecting gravity, which of the following are the correct differential equations of motion for the particle? d^2x/dt^2 , d^2y/dt^2 , d^2z/dt^2

- A. $d^2x/dt^2 = 0$, $d^2y/dt^2 = 0$, $d^2z/dt^2 = 0$
- B. $d^2x/dt^2 = -2x$, $d^2y/dt^2 = 4y$, $d^2z/dt^2 = 6z$
- C. $d^2x/dt^2 = 2x$, $d^2y/dt^2 = -4y$, $d^2z/dt^2 = 6z$
- D. $d^2x/dt^2 = -2x$, $d^2y/dt^2 = 4y$, $d^2z/dt^2 = -6z$ ✓

EXPLANATION

Newton's second law: $F_x = m \cdot d^2x/dt^2 = -4k \cdot x$.

40. A point particle is subject to a potential energy: $U(x,y,z) = (x^2 + y^2 + z^2) + mgz^2$ where k is a constant, m is the particle's mass, and g is gravitational acceleration. Which of the following are the correct motion equations?

- A. $x'' + kx = 0$, $y'' + ky = 0$, $z'' + kz = g m$
- B. $x'' - kx = 0$, $y'' - ky = 0$, $z'' - kz = -g m$
- C. $x'' - kx = 0$, $y'' - ky = 0$, $z'' - kz = g m$
- D. $x'' + kx = 0$, $y'' + ky = 0$, $z'' + kz = -g m$ ✓

EXPLANATION

Force from potential: $F_x = -dU/dx = -kx - m \cdot g$.

41. A point particle moves along a curvilinear path. Is it possible that its acceleration is zero?

- A. Yes, but only if its vector speed is constant.
- B. No, because there is always at least tangential acceleration.
- C. Yes, but only if its scalar speed is constant.
- D. No, because there is always at least centripetal acceleration. ✓

EXPLANATION

Acceleration is zero only if velocity is constant and straight.

42. A point particle of mass M moves in an inertial reference system along the positive x -axis. It is subjected to a braking force that depends on the speed according to the law: $F = -kv^2$ where k is a positive constant. Given that it passes through the origin with speed v at time: $t = 0$ determine the expressions for the speed $v(t)$ and the position $x(t)$ as functions of time t .

- A. $v(t) = -t^3 + v_0$, $x(t) = -t^4 + v_0 t / 12M$
- B. $v(t) = e^{-kt}$, $x(t) = \log(kt) + v_0 / k$ ✓
- C. $v(t) = v_0 + v_0 t/k$, $x(t) = v_0 t + v_0 t^2 / 2k$
- D. $v(t) = \sin(t) + v_0$, $x(t) = \sqrt{(\sin(t))} + v_0 t / 3M$

EXPLANATION

Velocity: $v(t) = v_0/(1 + (k \cdot v_0 \cdot t)/M)$.

43. A reversible thermal machine works with four heat sources. From the first source at temperature $T = 500\text{K}$, the machine absorbs heat $Q = 5000\text{J}$. At the fourth source, at temperature $T_4 = 280\text{K}$, the machine transfers heat $Q = -1400\text{J}$. With the second and third sources, at $T_2 = 400\text{K}$ and $T_3 = 300\text{K}$, the machine exchanges heat Q and $Q = -Q_2$ and Q_3 . Calculate the efficiency of the thermal machine.

- A. 0.127
- B. 0.327 ✓
- C. 0.427
- D. 0.227

EXPLANATION

Efficiency: $\eta = 1 - (Q_2/T_2 + Q_3/T_3 + Q_1/T_1 + Q_4/T_4)$.

44. A rod with mass m and length $L = 1.2\text{m}$ is hinged from one end to a wall. The rod is initially in a horizontal position and released from rest. What will its angular velocity ω be once it reaches the vertical position?

- A. $\omega = 5\text{s}^{-1}$ ✓
- B. $\omega = 15\text{s}^{-1}$
- C. $\omega = 0.5\text{s}^{-1}$
- D. $\omega = 0.25\text{s}^{-1}$

EXPLANATION

Energy conservation: $\frac{1}{2}mL^2\omega^2 = mgL/2 \Rightarrow \omega$

45. A simple pendulum used to measure time is made of a mass attached to the end of a steel wire. The period of the pendulum correctly shows 1 s when the temperature is 0°C . Calculate how much time the clock will lag or gain in 1 day in a hot country where the average temperature is 40°C . (The linear expansion coefficient of steel is $\alpha = 1.6 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$)

- A. The pendulum lags by 20 s per day ✓
- B. The pendulum lags by 2.0 s per day
- C. The pendulum lags by 10 s per day
- D. The pendulum lags by 1.0 s per day

EXPLANATION

$\Delta t = (\alpha \cdot \Delta T \cdot t)/2 \approx 20 \text{ s/day}$ ($\alpha = 1.6 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$, $\Delta T = 40^\circ\text{C}$, $t = 86400 \text{ s}$).

46. A small object is dropped to Earth from a great distance (e.g., of the order of the Earth-Moon distance).

- A.** does not describe a uniformly accelerated motion given the large distance from the Earth ✓
- B.** nothing can be said unless the mass of the object is known
- C.** describes a uniformly accelerated motion like any other body that falls to Earth from much
- D.** describes a uniform rectilinear motion

EXPLANATION

Gravity weakens with distance, so acceleration isn't constant (not uniformly accelerated).

47. A small (glass) sphere of radius r and density ρ , initially at rest, is dropped into a container full of water of density ρ_0 and viscosity η . Write the differential equation of motion for the small sphere.

- A.** $d^2x(t)/dt^2 + 9\eta/2r^2 (dx(t))/dt = (1-\rho_0/\rho)g$
- B.** $d^2x(t)/dt^2 - 9\eta/2r^2 (dx(t))/dt = (1-\rho_0/\rho)g$
- C.** $d^2x(t)/dt^2 + 9\eta/2r^2 (dx(t))/dt = (1-\rho_0/\rho)g$ ✓
- D.** $d^2x(t)/dt^2 - 9\eta/2r^2 (dx(t))/dt = -(1-\rho_0/\rho)g$

EXPLANATION

Force balance: $m \cdot d^2x/dt^2 = -mg + 6\pi\eta r \cdot dx/dt + (\rho - \rho_0)g \cdot (4/3)\pi r^3$.

48. A solid sphere with mass M and radius R is welded to the end of a rod with mass m and length L . Calculate the moment of inertia I of this system with respect to the y -axis at the opposite y end of the rod.

- A.** $I = \frac{1}{3} mL^2 + \frac{2}{5} MR^2$
- B.** $I = \frac{1}{3} mL^2 + \frac{2}{5} MR^2$
- C.** $I = \frac{1}{3} mL^2 + \frac{2}{5} MR^2 + M(L+R)^2$ ✓
- D.** $I = M(L+R)^2$

EXPLANATION

$I = I_{\text{rod}} + I_{\text{sphere}} = \frac{1}{3} mL^2 + \frac{2}{5} MR^2 + M(L+R)^2$.

49. A sphere with radius $R = 0.25\text{m}$ and mass $M = 1\text{kg}$ is released on an inclined plane from a height $h = 10.5\text{m}$ above the ground and rolls down without slipping. What will the velocity of the center of mass be once it reaches the ground?

- A.** 12 m/s ✓
- B.** 4.0 m/s
- C.** 1.2 m/s
- D.** 24.0 m/s

EXPLANATION

Energy conservation: $mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$; $I = \frac{2}{5}mR^2$, $\omega = v/R \rightarrow v = \sqrt{(10gh/7)} \approx 12\text{ m/s}$.

50. A sphere of 10 g hits centrally and elastically, at a speed of 5 m/s, another stationary sphere of unknown mass. After the impact, it goes back with a speed of $2 \frac{\text{m}}{\text{s}}$. Is it possible, neglecting frictions, to calculate the mass of the impacted sphere?

- A. No, a collision without friction is nonsense
- B. No, because in this type of impact the total energy of the system composed of the two
- C. No, because the speed of the second sphere after the impact is unknown
- D. Yes ✓

EXPLANATION

Yes, using conservation of momentum and kinetic energy (elastic collision).

51. A student is trying to decide what to wear. The surroundings (his bedroom) are at 20.0°C . If the skin temperature of the unclothed student is 35°C , what is the net energy loss from his body in 10.0 min by radiation? Assume the emissivity of skin is 0.900 and the surface area of the student is 1.50 m^2 .

- A. $Q = 7.5 \cdot 10^3 \text{ J}$
- B. $Q = 7.5 \cdot 10^5 \text{ J}$
- C. $Q = 7.5 \cdot 10^2 \text{ J}$
- D. $Q = 7.5 \cdot 10^4 \text{ J}$ ✓

EXPLANATION

$Q = \epsilon \sigma A (T_1^4 - T_2^4) t \approx 7.5 \text{e}4 \text{ J}$ ($\sigma = 5.67 \text{e-}8 \text{ W/m}^2\text{K}^4$).

52. A system performs a thermodynamic cycle. Its entropy variation

- A. is always positive
- B. is null if the cycle is reversible
- C. depends on the thermodynamic cycle, it is positive or negative
- D. is null ✓

EXPLANATION

Entropy is a state function; over a cycle, $\Delta S = 0$.

53. A thermal machine operates with three heat sources. T_1 , T_2 , and T_3 are the temperatures of the 1 2 3 sources, and Q_1 , Q_2 , Q_3 are the amounts of heat that the thermal machine exchanges with each 1 2 3 of the sources. If the cycle of the thermal machine is irreversible, then it is always true that:

- A. $Q_1 + Q_2 + Q_3 < 0$ ✓
- B. $Q_1/T_1 + Q_2/T_2 + Q_3/T_3 = 0$
- C. $Q_1 + Q_2 + Q_3 > 0$
- D. $Q_1 + Q_2 + Q_3 = 0$

EXPLANATION

For irreversible cycles, total entropy change of the universe is negative ($Q_1 + Q_2 + Q_3 < 0$).

54. A tire, which is initially at the temperature $T = 300\text{K}$, is inflated to a pressure equal to three **times** that of the initial pressure. Assuming that this process is adiabatic and almost static, determine the final temperature T of the air considered as a perfect diatomic gas.

- A. $T = 300\text{K}$
- B. $T = 100\text{K}$
- C. $T = 411\text{K}$ ✓
- D. $T = 900\text{K}$

EXPLANATION

C) $T = 411\text{K}$, using $T_2 = T_1(P_2/P_1)^{((\gamma-1)/\gamma)}$, $\gamma = 1.4$ for diatomic gas.

55. A vessel contains 4 moles of a monatomic gas at $T = 0^\circ\text{C}$. The temperature of the gas is increased to $T = 50^\circ\text{C}$ at constant volume. Calculate the given heat Q , the work performed W by the gas, and the increase in internal energy U .

- A. $Q = 2500\text{J}$, $W = 0\text{J}$, $\Delta U = 0\text{J}$ ✓
- B. $Q = 2500\text{J}$, $W = 2500\text{J}$, $\Delta U = 2500\text{J}$
- C. $Q = 2500\text{J}$, $W = 2500\text{J}$, $\Delta U = 0\text{J}$
- D. $Q = 2600\text{J}$, $W = 0\text{J}$, $\Delta U = 2500\text{J}$

EXPLANATION

$Q = nC_v\Delta T = 4 \cdot (3R/2) \cdot 50 \approx 2500\text{ J}$; $W=0$ (constant volume); $\Delta U=Q$.

56. A vessel contains $V = 10\text{L}$ of ideal gas at pressure $P = 5\text{atm}$. This gas is first cooled at a constant volume along the path $a \rightarrow b$ until its pressure reaches $P = 2\text{atm}$. Then, it is again brought to temperature T by expansion under constant pressure ($b \rightarrow c$). What is the volume at $V_{(c)}$? What is the total work W performed by the gas? What is the total increase in internal energy U ?

- A. $V = 2.5\text{L}$; $W = 300\text{J}$; $U = 0\text{J}$
- B. $V = 25\text{L}$; $W = 0\text{J}$; $U = 0\text{J}$
- C. $V = 25\text{L}$; $W = 3000\text{J}$; $U = 0\text{J}$
- D. $V = 25\text{L}$; $W = 3000\text{J}$; $U = 3000\text{J}$ ✓

EXPLANATION

$V_{(c)} = V_{(b)} \cdot ((P_{(b)})/(P_{(c)})) = 25\text{ L}$; $W = P_{(b)}(V_{(c)}-V_{(b)}) = 3000\text{ J}$; $\Delta U = Q = W$ (isobaric).

57. A wheel of radius R rotates around a fixed axis. When a point at a distance R from the center moves with an angular speed ω , a second point, located at a distance $R/2$ from the center, moves with an angular velocity:

- A. ω ✓
- B. 2ω
- C. $\omega/2$
- D. 4ω

EXPLANATION

A) ω : Angular velocity is the same for all points on a rigid rotating body.

58. Air density is 1.3 kg/m^3 at sea level. What would the thickness of the Earth's atmosphere be if the density of air did not decrease with height, remaining constant?

- A. 100 km
- B. 78 km
- C. 7.8 km ✓
- D. Not enough information to say

EXPLANATION

C) 7.8 km: Atmospheric thickness = $P/(\rho g) = 101325/(1.3 \times 9.81) \approx 7.9 \text{ km}$.

59. Air undergoes a three-process cycle with a $P = \text{const}$ process, a $T = \text{const}$ process, and a $V = \text{const}$ process. Select the correct statement for a piston-cylinder arrangement.

- A. $W = 0$ for the $P = \text{const}$ process
- B. $Q = 0$ for the $V = \text{const}$ process
- C. $Q = 0$ for the $T = \text{const}$ process
- D. $W = 0$ for the $V = \text{const}$ process ✓

EXPLANATION

D) $W = 0$ for the $V = \text{const}$ process: Work $W = \int P dV = 0$ when volume is constant.

60. An aerodynamic tunnel must be used with a model car of size 20 cm to roughly reproduce the situation in which a car of size 550 cm travels at 15 m/s. What should be the wind speed in the tunnel? (Reynolds number has to be the same.)

- A. 411 m/s ✓
- B. (C)
- D. To match the Reynolds number:

EXPLANATION

A) 411 m/s: Reynolds number $Re = \rho vL/\eta$ must be equal $\rightarrow v_{\text{model}} = v_{\text{prototype}} \times (L_{\text{prototype}}/L_{\text{model}})$.

61. An aluminum block of mass $m = 0.1 \text{ kg}$ at temperature $T = 580 \text{ K}$ is immersed in a glass vessel of mass $m = 0.2 \text{ kg}$ and having temperature $T = 300 \text{ K}$. The glass vessel contains $m = 0.5 \text{ kg}$ of water at $T = 300 \text{ K}$. Disregarding heat exchange with the surroundings, determine the equilibrium temperature T of the system. The specific heats are: $c_1 = 896.0 \text{ J/kg} \cdot \text{K}$ (aluminum), $c_2 = 630.4 \text{ J/kg} \cdot \text{K}$ (glass), $c_3 = 4187.0 \text{ J/kg} \cdot \text{K}$ (water)

- A. $T = 410.86 \text{ K}$
- B. $T = 440.00 \text{ K}$
- C. $T = 310.86 \text{ K}$ ✓
- D. $T = 390.66 \text{ K}$

EXPLANATION

C) $T = 310.86 \text{ K}$, from heat balance: $m_1 c_1 (T_1 - T) = m_2 c_2 (T - T_2) + m_3 c_3 (T - T_2)$.

62. An ice cube (density $\rho = 920.0 \text{ kg/m}^3$) with edge $a = 4.0 \text{ cm}$ floats in a cylindrical glass with ice section area $S = 25.00 \text{ cm}^2$, filled with water up to a height $h = 10.0 \text{ cm}$. Calculate the new height h' of the water when the cube is completely melted.

- A. $h' = 11.0 \text{ cm}$
- B. $h' = 9.0 \text{ cm}$
- C. $h' = 10.0 \text{ cm}$ ✓
- D. $h' = 9.5 \text{ cm}$

EXPLANATION

C) $h' = 10.0\text{cm}$, volume displaced by ice = volume added by melted ice.

63. An inventor claims that a thermal engine, operating between ocean layers at 27°C and 10°C , produces 10kW of power while discharging 9900kJ/min . This engine is:

- A.** impossible ✓
- B.** reversible
- C.** possible
- D.** probable

EXPLANATION

A) impossible: The claimed efficiency exceeds the Carnot limit ($\eta = 1 - 283/300 \approx 5.7\%$).

64. An object oscillates with simple harmonic motion along the x-axis. Its displacement from the origin varies with time according to the equation: $x(t) = (4.0\text{m})\cos(\pi t + \pi/4)$ where t is in seconds and the angles are in radians. Determine the amplitude A , frequency f , and period T of the motion.

- A.** $A = 4.0\text{m}$; $f = 0.5\text{Hz}$; $T = 2.0\text{s}$ ✓
- B.** $A = 4.0\text{m}$; $f = 1.5\text{Hz}$; $T = 0.7\text{s}$
- C.** $A = 4.0\text{m}$; $f = 0.25\text{Hz}$; $T = 4.0\text{s}$
- D.** $A = 40\text{m}$; $f = 0.5\text{Hz}$; $T = 2.0\text{s}$

EXPLANATION

A) $A = 4.0\text{m}$; $f = 0.5\text{Hz}$; $T = 2.0\text{s}$: From $x(t) = A \cos(\pi t + \pi/4)$, $\omega = \pi \text{ rad/s} \rightarrow f = \omega/(2\pi) = 0.5 \text{ Hz}$, $T = 1/f = 2.0 \text{ s}$.

65. An oscillator has an elastic constant k , mass m , and is subjected to a braking force proportional to the speed ($F = -\lambda v$). It performs damped oscillations just when:

- A.** $\lambda^2 < 4mk$ ✓
- B.** $\lambda^2 > 4mk$
- C.** $\lambda^2 < mk$
- D.** $\lambda^2 > mk$

EXPLANATION

A) $\lambda^2 < 4mk$: Damped oscillations occur when damping coefficient $\lambda < 2\sqrt{(mk)}$.

66. Bernoulli's theorem:

- A.** claims that in the points of a duct where the section is smaller, vortices are produced that
- B.** deals with the relationship between speed and pressure in a fluid at different points in the
- C.** deals with the relationship between speed and pressure in a fluid at different points in the same conduct ✓
- D.** deals with the relationship between the flow rate of a conduct and the viscous friction

EXPLANATION

C) deals with the relationship between speed and pressure in a fluid at different points in the same conduct: Bernoulli's theorem states $P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$ along a streamline.

67. By introducing in a calorimeter at room temperature 300 g of water at 353 K, it is observed that at equilibrium the temperature is 323 K. Is it possible to deduce from this the thermal capacity of the calorimeter?

- A. No, other data are required
- B. No, you need to know the nature of the various parts that make up the calorimeter, their
- C. Yes ✓
- D. No, is not an exact differential

EXPLANATION

C) Yes: Calorimeter heat capacity can be deduced from temperature changes of water and calorimeter.

68. Calculate the moment of inertia of a system consisting of the three masses shown in the figure with respect to each of the x, y, and z axes.

- A. $I = 2.0\text{kg}\cdot\text{m}^2$; $I = 4.8\text{kg}\cdot\text{m}^2$; $I = 6.8\text{kg}\cdot\text{m}^2 \times y \ z$
- B. $I = 20\text{kg}\cdot\text{m}^2$; $I = 4.8\text{kg}\cdot\text{m}^2$; $I = 68\text{kg}\cdot\text{m}^2 \times y \ z$
- C. $I = 20\text{kg}\cdot\text{m}^2$; $I = 48\text{kg}\cdot\text{m}^2$; $I = 68\text{kg}\cdot\text{m}^2 \times y \ z$ ✓
- D. $I = 20\text{kg}\cdot\text{m}^2$; $I = 48\text{kg}\cdot\text{m}^2$; $I = 6.8\text{kg}\cdot\text{m}^2 \times y \ z$

EXPLANATION

C) $I_x = \sum m_i r_i^2 = 20\text{kg}\cdot\text{m}^2$, $I_y = 48\text{kg}\cdot\text{m}^2$, $I_z = 68\text{kg}\cdot\text{m}^2$.

69. Calculate the total entropy change if 10kg of ice at 0°C is mixed in an insulated container with 20kg of water at 20°C. (The heat of fusion for ice is 340kJ/kg, and the process is thermally isolated.)

- A. 6.1kJ/K
- B. 3.9kJ/K
- C. 1.2kJ/K
- D. 0.21kJ/K ✓

EXPLANATION

D) 0.21 kJ/K: Total entropy change = entropy gained by ice + entropy lost by water.

70. Clapeyron's formula, for phase transitions, states that the derivative of pressure versus temperature, dP/dT

- A. Directly proportional to the temperature: Incorrect. It is inversely proportional to T —
- B. Inversely proportional to the latent heat: Incorrect. It is directly proportional to L, not inversely.
- C. Does not depend on the density of the substance: Incorrect. Since ΔV involves volume and
- D. Inversely proportional to the temperature of the substance: Correct, as seen from the ✓

EXPLANATION

D) Inversely proportional to the temperature of the substance: Correct, as seen from the Clapeyron equation $dP/dT = L/(T\Delta V)$.

71. Consider $n = 0.20$ moles of nitrogen (a diatomic gas) initially at temperature $T = 300\text{K}$ and 1 pressure $P = 105\text{Pa}$. The gas performs a reversible adiabatic compression until the pressure reaches $P = 2P$. Calculate the final volume V , the final temperature T , and the work $W_{(F)_i}$ done by the gas.

- A. $V = 3.0 \cdot 10^{-3}\text{m}^3$; $T = 300\text{K}$; $W = -254\text{J}$
- B. $V = 3.0 \cdot 10^{-3}\text{m}^3$; $T = 329\text{K}$; $W = -254\text{J}$
- C. $V = 3.0 \cdot 10^{-3}\text{m}^3$; $T = 361\text{K}$; $W = -254\text{J}$ ✓
- D. $V = 3.0 \cdot 10^{-3}\text{m}^3$; $T = 361\text{K}$; $W = -154\text{J}$

EXPLANATION

C) $V = 3.0 \cdot 10^{-3}\text{m}^3$, $T = 361\text{K}$, $W = -nC_v(T_2 - T_1) = -254\text{J}$.

72. Determine the final temperature t of a mixture when a mass $m = 100\text{g}$ of ice (fusion heat $l_f = 80\text{cal/g}$) at temperature $t_i = 0^\circ\text{C}$ is mixed with a mass $m = 500\text{g}$ of water at temperature $t_f = 50^\circ\text{C}$, assuming no heat loss to the environment. Given: specific heat of water $c = 2 \text{ w } 1\text{cal/g} \cdot ^\circ\text{C}$

- A. $t = 28.33^\circ\text{C}$ ✓
- B. $t = 38.33^\circ\text{C}$
- C. $t = 25.00^\circ\text{C}$
- D. $t = 18.33^\circ\text{C}$

EXPLANATION

A) $t = 28.33^\circ\text{C}$: Heat lost by water = heat gained by ice ($Q = mc\Delta T + mL + mc\Delta T$).

73. Determine the speed v that a homogeneous sphere reaches at the end of its descent on an inclined plane, starting from a height $h = 1.14\text{m}$, assuming it rolls without slipping.

- A. $v = 4\text{m/s}$ ✓
- B. $v = 0.4\text{m/s}$
- C. $v = 40\text{m/s}$
- D. The mass and radius of the sphere are not known therefore it is impossible to answer

EXPLANATION

A) $v = 4 \text{ m/s}$: Energy conservation: $mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \rightarrow v = \sqrt{(4gh/3)}$.

74. Does the second law of thermodynamics also apply to electromagnetic phenomena?

- A. Yes. ✓
- B. No, since the efficiency of electric motors is higher than that of heat engines.
- C. Yes, but only for static electric or magnetic phenomena.
- D. It depends, it is necessary to specify case by case.

EXPLANATION

A) Yes.: The second law applies to all thermodynamic processes, including electromagnetic.

75. During the laying of a high voltage pylon, a bolt falls from a height h above the ground. It is known that in the last second of its motion before hitting the ground, it travels a height equal to $h/2$. Determine the value of h .

- A. $h = 37.2\text{m}$
- B. $h = 47.2\text{m}$
- C. $h = 67.2\text{m}$
- D. $h = 57.2\text{m}$ ✓

EXPLANATION

D) $h = 57.2$ m: Use kinematic equations for free fall to solve for height h .

76. Find the work needed to compress 2kg of air in an insulated cylinder from 100kPa to 600kPa if $T = 20^\circ\text{C}$ (air molecular weight = 28.96g/mol).

- A. -469kJ
- B. -390kJ
- C. -280kJ
- D. -220kJ ✓

EXPLANATION

D) -220 kJ: Work for adiabatic compression $W = nC_v\Delta T = (m/M)C_v(T_2 - T_1)$.

77. For a system of particles on which no external forces act, it is always true that:

- A. the velocity vector of the center of mass of the system is necessarily zero.
- B. the velocity vector of the center of mass of the system is constant. ✓
- C. the vector momentum of the center of mass of the system is necessarily zero.
- D. the vector angular momentum of the system with respect to the center of mass is necessarily zero.

EXPLANATION

B) the velocity vector of the center of mass of the system is constant.: For an isolated system, center of mass velocity is constant.

78. For a rigid body, initially at rest, to be in static equilibrium:

- A. it is sufficient that the resultant of the external forces is zero.
- B. it is sufficient that the resulting moment of external forces with respect to the center of mass is zero.
- C. it is necessary that the resultant of external forces is zero (net force and net torque both zero).. ✓
- D. it is sufficient that all external forces are conservative.

EXPLANATION

C) it is necessary that the resultant of external forces is zero (net force and net torque both zero).: Static equilibrium requires both net force and net torque to be zero.

79. For the second law of thermodynamics:

- A. the efficiency of a thermal machine is independent of the substance that completes the cycle
- B. the efficiency of a Carnot machine is as low as possible
- C. all thermal machines have the same efficiency
- D. the efficiency of a Carnot machine depends only on the temperature of the sources ✓

EXPLANATION

D) the efficiency of a Carnot machine depends only on the temperature of the sources: Carnot efficiency $\eta = 1 - T_{\text{Cold}}/T_{\text{hot}}$.

80. Four moles of an ideal gas undergo expansion from volume V to $V = 3.45V$. If the expansion is isothermal at temperature $T = 410\text{K}$, find the change in entropy ΔS of the gas. If instead the expansion is a reversible adiabatic process, what is the entropy change ΔS of the gas?

- A. $\Delta S = -41.2\text{J/K}$; $\Delta S = 0\text{J/K}$
- B. $\Delta S = 0\text{J/K}$; $\Delta S = 0\text{J/K}$
- C. $\Delta S = 41.2\text{J/K}$; $\Delta S = -41.2\text{J/K}$
- D. $\Delta S = 41.2\text{J/K}$; $\Delta S = 0\text{J/K}$ ✓

EXPLANATION

D) $\Delta S = 41.2\text{ J/K}$; $\Delta S = 0\text{ J/K}$: For isothermal expansion $\Delta S = nR \ln(V_2/V_1)$; for adiabatic $\Delta S = 0$.

81. Frictional forces (e.g., due to motion in a fluid, air, water, etc.) are generally not conservative. Can they be conservative in some particular cases?

- A. Yes, if the motion is laminar and no vortices are formed.
- B. Yes, if you neglect the form factor.
- C. No, in no case. ✓
- D. Yes, if the forces are strictly proportional to the speeds.

EXPLANATION

C) No, in no case.: Frictional forces are always non-conservative due to energy dissipation.

82. Frictional forces (e.g., due to motion in a fluid, air, water, etc.) are generally not conservative. Can they be conservative in some particular cases?

- A. Yes, if the motion is laminar and no vortices are formed.
- B. Yes, if you neglect the form factor.
- C. No, in no case. ✓
- D. Yes, if the forces are strictly proportional to the speeds.

EXPLANATION

Frictional forces dissipate energy as heat, so they cannot be conservative., Q2: $Re = \rho v(2r)/\eta = 10^3 \cdot 1 \cdot 0.2 / 10^{-3} = 2 \times 10^5 > 2300 \rightarrow$ turbulent., Q3: Negative slope implies $dT/dP < 0$, so melting point decreases with pressure., Q4: $Re = \rho v(2r)/\eta = 997 \cdot 10^{-3} \cdot 1 / 8.9 \times 10^{-4} \approx 1120 < 2300 \rightarrow$ laminar., Q5: Mayer's relation generalizes to $C_p - C_v = R$, allowing C to vary with temperature., Q6: Hydrostatic equilibrium gives $dp/dz = -\rho g$, solved as $p(z) = p(0)\exp(-z/a)$., Q7: $Q = mc\Delta T + mL + mc\Delta T = 50 \cdot 0.5 \cdot 30 + 50 \cdot 80 + 50 \cdot 1 \cdot 20 = 5750$ cal., Q8: Equipartition gives $E = (3/2)kT$ for monatomic gas., Q9: $\nabla \times F = (0, -5c, 0) \neq 0$ unless $a = b = 0 \rightarrow$ conservative only then., Q10: $\nabla \times F = (0, 0, 0)$ only if $a = b = 0 \rightarrow$ never conservative., Q11: Internal energy of ideal gas depends only on temperature ($U = (3/2)nRT$ for monatomic)., Q12: Both U and S are state functions, depending only on system state., Q13: $C_v = (11/2)R \rightarrow f = 11 \rightarrow$ polyatomic (e.g., nonlinear molecule)., Q14: Moment of inertia is a property of mass distribution, independent of motion., Q15: Resonance occurs at natural frequency ω_0 when driving frequency $\Omega = \omega_0$., Q16: Van der Waals equation corrects ideal gas law with attraction (a) and excluded volume (b)., Q16: $W = \int F \cdot dr = k \int (xdx + ydy + zdz) = k(x^2 + y^2 + z^2)/2 = 2(1+0+1)/2 = 2$ J., Q18: $W = -k \int (xdx + ydy + zdz) = -k(x^2 + y^2 + z^2)/2 = -2(4+1+4)/2 = -9$ J (magnitude 9 J)., Q19: $\nabla \times F = (0, 0, 0)$ always $\rightarrow$ conservative for any a, b, c ., Q20: $\nabla \times F = 0$ for central forces $\rightarrow$ conservative., Q21: Hagen-Poiseuille applies to incompressible laminar flow in cylindrical tubes., Q22: Valid only for laminar flow ($Re < 2300$) in cylindrical ducts., Q23: In adiabatic process, $\Delta U = Q + W \rightarrow W = -\Delta U = -2$ J (work done on system = 2 J)., Q24: Moment of inertia depends on axis orientation relative to mass distribution., Q25: Mass is invariant; weight changes due to reduced gravitational acceleration at height., Q26: $PV = \text{const}$ is isothermal, $PV^\gamma = \text{const}$ is adiabatic $\rightarrow$ cannot both hold unless $\gamma = 1$ (invalid)., Q27: Use isentropic relations for steam: $W \approx nC_v(T_1 - T_2) \approx 940$ kJ., Q28: For irreversible cycles, total entropy change $\Delta S_{\text{total}} < 0$ (Clausius inequality)., Q29: Weight = mg is independent of contact area or center of gravity position., Q30: Time of flight depends only on vertical motion: $t = 2v_0 \sin \theta / g$ (mass cancels)., Q31: Mixing increases entropy ($\Delta S > 0$) in isolated system., Q32: Orbital period $T = 2\pi\sqrt{r^3/GM} \rightarrow$ Earth's higher gravity gives shorter period., Q33: Stirling efficiency depends only on source temperatures ($\eta = 1 - T_{\text{cold}}/T_{\text{hot}}$)., Q34: Temperature is directly proportional to average kinetic energy ($T = (2/3k)\langle KE \rangle$)., Q35: Both relations derive from ideal gas law and $\gamma = C_p/C_v \rightarrow$ statement

83. From Mayer's relation for the specific heats of ideal gases $C_p - C_v = R$ it can be deduced in a more general way that:

- A. is incorrect — C differs between gases.
- B. is incorrect — C is not always constant.
- C. is incorrect — it does depend on temperature.
- D. correctly states that C can be a function of temperature and under certain conditions ✓

EXPLANATION

Mayer's relation $C_p - C_v = R$ holds generally, with C_p and C_v possibly temperature-dependent.

84. How does the pressure p in the atmosphere change with altitude z , assuming the atmosphere behaves as a perfect gas? (Here, a is a constant.)

- A. $p(z) = p(z=0)gz$, where ρ is the air density
- B. $p(z) = p(z=0)\exp(-z/a)$ ✓
- C. $p(z) = p(z=0)\exp(z/a)$
- D. $p(z) = p(z=0)z/a$

EXPLANATION

Pressure decreases exponentially with altitude: $p(z) = p_0 \exp(-z/a)$.

85. How much heat Q is required to turn a block of ice with mass $m = 50\text{g}$ at temperature $T = -30^\circ\text{C}$ into water at $T = 20^\circ\text{C}$? Latent heat of fusion: $L = 80\text{cal/g}$ Specific heat of ice: $1\text{ w}_\text{(c)}$ $= 0.5\text{cal/g}^\circ\text{C}$ Specific heat of water: $c = 1\text{cal/g}^\circ\text{C}$

- A.** $Q = 5750\text{ cal}$ ✓
- B.** $Q = 575\text{ cal}$
- C.** $Q = 6\text{ cal}$
- D.** $Q = 58\text{ cal}$

EXPLANATION

$Q = mc\Delta T + mL + mc\Delta T = 50 \times 0.5 \times 30 + 50 \times 80 + 50 \times 1 \times 20 = 5750\text{ cal}.$

86. How much is the average energy E of a molecule of an ideal gas of monoatomic molecules at temperature T ?

- A.** $E = 3/2kT$ ✓
- B.** $E = 4kT$
- C.** $E = 5kT$
- D.** $E = kT$

EXPLANATION

Average energy per molecule in ideal gas: $E = 3/2kT.$

87. How much is the gravitational field F and the gravitational potential energy V produced by a homogeneous spherical mass M of radius R at a point inside the sphere, at a distance $r < R$ from the center? (Here, C is an integration constant.) GMr

- A.** $F = -GMr/r^3, V = +C$
- B.** $F = -GM/r^2, V = +C$ ✓
- C.** $F = -GM/r^2, V = -GM/r + C$
- D.** $F = -GM/r^2, V = -GM/r + C$

EXPLANATION

$F = -GMm/r^3, V = -GMm/r + C.$

88. How much is the gravitational force F and energy U exerted between two homogeneous spherical masses M_1 and M_2 , if the distance between their centers is r , with r greater than the radii of each mass?

- A.** $F = -GM_1M_2/r^3; U = -GM_1M_2/r$
- B.** $F = -GM_1M_2/r^2; U = -GM_1M_2/r^2$
- C.** $F = -GM_1M_2/r^2; U = -GM_1M_2/r$ ✓
- D.** $F = -GM_1M_2/r^3; U = -GM_1M_2/r^2$

EXPLANATION

C) $F = -GM_1M_2/r^2; U = -GM_1M_2/r$: Newton's law of gravitation and gravitational potential energy.

89. How will a pendulum clock made of metal wire show the time when temperature increases?

- A.** It will lag. ✓
- B.** It will run too fast.
- C.** It will not change.
- D.** It is impossible to tell.

EXPLANATION

A) It will lag.: Pendulum period increases with length,

90. If a right horizontal tube of radius $r = 0.1\text{m}$ has water moving with speed $v = 1\text{m/s}$, is the flow laminar or turbulent? (Given: dynamic viscosity $\eta = 10^{-3}\text{ kg/(m}\cdot\text{s)}$, water density $\rho = 10^3\text{ kg/m}^3$)

- A. Laminar
- B. The data of this question are insufficient to answer
- C. It is necessary to also know the water temperature to answer
- D. Turbulent ✓

EXPLANATION

Turbulent flow occurs when Reynolds number exceeds ~ 2000 , here $Re = \rho v r / \eta = (10^3 \times 1 \times 0.1) / (10^{-3}) = 100,000 > 2000$.

91. If $a = -kv^2$, where k is a constant, determine $v(t)$ and $x(t)$ given the initial conditions: $x(0) = 0$ and $v(0) = v_0$

- A. $x(t) = \ln(v_0 k t - 1)$, $v(t) = v_0 k v_0 k t + 1$
- B. $x(t) = \ln(v_0 k t + 1)$, $v(t) = v_0 k v_0 k t + 1$ ✓
- C. $x(t) = \ln(v_0 k t + 1)$, $v(t) = v_0 k (v_0 k t + 1)$
- D. $x(t) = e^{-kt} - t + 1$, $v(t) = e^{-kt} - 1 k k dv/dt$

EXPLANATION

B) $x(t) = \ln(v_0 k t + 1)/k$, $v(t) = v_0 / (v_0 k t + 1)$, solving $dv/dt = -kv^2$.

92. In a PT (Pressure-Temperature) diagram, the line that separates the solid phase from the liquid phase of water has a negative slope. This implies that:

- A. The melting temperature of the ice decreases with increasing pressure. ✓
- B. The water decreases in volume when it freezes.
- C. The melting temperature of the ice is always zero degrees Celsius.
- D. The melting temperature of the ice increases with increasing pressure.

EXPLANATION

Negative slope in PT diagram means melting point decreases with pressure.

93. In a horizontal pipe with a diameter of 1m , water flows at a speed of 10^{-3} m/s . Is the regime laminar? (The water viscosity is $\eta = 8.9 \times 10^{-4}\text{ Pa}\cdot\text{s}$, and the water density is $\rho = 997\text{ kg/m}^3$.)

- A. Yes ✓
- B. No
- C. The data are insufficient to answer the question
- D. In a horizontal tube they are always in laminar regime

EXPLANATION

$Re = \rho v d / \eta = 997 \times 10^{-3} \times 1 / 8.9 \times 10^{-4} \approx 1120 < 2000$, so laminar.

94. In a region of space, the potential energy is given by $U(x,y,z) = b(x^2 + y^2 + z^2) - bxy + c$ where b and c are constants of suitable size. What is the analytical expression of the force $\vec{F}(x,y,z)$?

- A. $\vec{F}(x,y,z) = b(-2x+xy)\hat{i} + b(-2y+xy)\hat{j} - 2bz\hat{k}$
- B. $\vec{F}(x,y,z) = b(2x-y)\hat{i} + b(2y-x)\hat{j} + (2bz+c)\hat{k}$
- C. $\vec{F}(x,y,z) = b(2x-y)\hat{i} + b(2y-x)\hat{j} + 2bz\hat{k}$
- D. $\vec{F}(x,y,z) = b(-2x+y)\hat{i} + b(-2y+x)\hat{j} - 2bz\hat{k}$ ✓

EXPLANATION

D) $\vec{F} = -\nabla U = b(-2x+y)\hat{i} + b(-2y+x)\hat{j} - 2bz\hat{k}$.

95. In a region of space, the potential energy is given by: $U(x,y,z) = 2ax^2 - 3axy + az^2$ where a is a constant of suitable size. What is the analytical expression of the force $\vec{F}(x,y,z)$?

- A. $\vec{F}(x,y,z) = -4ax\hat{i} + 3ay\hat{j} - 2az\hat{k}$
- B. $\vec{F}(x,y,z) = -4ax\hat{i} + 3ax\hat{j} - 2az\hat{k}$
- C. $\vec{F}(x,y,z) = a(3y-4x)\hat{i} + 3ax\hat{j} - 2az\hat{k}$ ✓
- D. $\vec{F}(x,y,z) = 4ax\hat{i} - 3ay\hat{j} + 2az\hat{k}$

EXPLANATION

$\vec{F} = -\nabla U = -[\partial U/\partial x, \partial U/\partial y, \partial U/\partial z] = a(3y-4x, 3x, -2z)$.

96. In a thermodynamical transformation, it is observed that a perfect gas doubles its volume. Therefore:

- A. its entropy variation is zero.
- B. there is not enough data to answer this question.
- C. its entropy variation is definitely negative.
- D. its entropy variation is certainly positive. ✓

EXPLANATION

Doubling volume at constant temperature increases entropy ($\Delta S > 0$).

97. In an infinitesimal adiabatic transformation, the work:

- A. is an exact differential
- B. is an exact differential only if the system temperature is close to 0 K
- C. is an exact differential only if it is a perfect gas
- D. is never an exact differential ✓

EXPLANATION

Work in adiabatic transformations is path-dependent, hence not an exact differential.

98. In any reversible transformation from a state A to a state B of a thermodynamic system, the entropy variation:

- A. depends only on states A and B ✓
- B. can only be calculated if the system is a perfect gas
- C. depends on the path taken to go from A to B
- D. is always different from zero

EXPLANATION

Entropy is a state function, depending only on initial and final states, not the path taken.

99. In the absence of friction, do wood pendulums continue to oscillate indefinitely?

- A. Yes, because the period of oscillation of the pendulum is constant.
- B. No, because the weight force is non-conservative.
- C. Yes, because the pendulum energy cannot dissipate. ✓
- D. No, because the motion of the pendulum is not a uniform motion.

EXPLANATION

Without friction, no energy dissipates, so oscillations continue indefinitely.

100. Is it possible to melt a piece of ice placed in an adiabatic vessel of strictly constant volume?

- A. Yes, for the first law of thermodynamics
- B. Yes, because ice has a high thermal capacity
- C. No, for the first law of thermodynamics ✓
- D. Yes, because the melting ice decreases in volume

EXPLANATION

No heat enters/leaves in an adiabatic vessel, so melting (requiring heat) is impossible.

101. Knowing that R is the tube radius, L the tube length, η is the fluid viscosity, and Δp is the pressure variation along the tube ends, the Hagen-Poiseuille equation for the volumetric flow rate q is: $q = (\pi R^2 \Delta p) / \dots$

- A. $q = -8\eta L^2$
- B. $q = -8L$
- C. $q = -8\eta L$ ✓
- D. $q = -8\eta L$

EXPLANATION

C) $q = (\pi R^4 \Delta p) / (8\eta L)$, Hagen-Poiseuille equation.

102. Knowing that the mass of a planet is $M = 5.98 \times 10^{26}$ kg and the radius is $R = 6.37 \times 10^6$ m, calculate the escape velocity v of a body from the planet.

- A. $v = 1.11 \times 10^3$ m/s
- B. $v = 1.11 \times 10^4$ m/s
- C. $v = 1.11 \times 10^5$ m/s ✓
- D. It is not possible to answer because the mass of the body is not known

EXPLANATION

$v = \sqrt{2GM/R} = \sqrt{(2 \cdot 6.67 \times 10^{-11} \cdot 5.98 \times 10^{26}) / (6.37 \times 10^6)} \approx 1.11 \times 10^5$ m/s.

103. Lead is denser than copper. When two equal masses of lead and copper are submerged in water, which one will experience a larger buoyant force?

- A. Lead
- B. Copper ✓
- C. Equal
- D. It is impossible to tell

EXPLANATION

Buoyant force = $\rho_{\text{water}} \cdot V_{\text{displaced}} \cdot g$; copper displaces more water (lower density).

104. Newton's second law of dynamics applied to a system of variable mass $m(t)$ is written as

- A. $F = m \cdot dv - v \cdot dm/dt$
- B. $F = (d(mv))/dt$ ✓
- C. $F = m \cdot dv/dt$
- D. $F = (d(m \cdot dv))/dt$

EXPLANATION

Newton's second law for variable mass is $F = d(mv)/dt$.

105. On a cold winter day, metallic objects generally feel cooler to the touch than wooden objects. This is because:

- A. metals conduct heat better than wood ✓
- B. the mass density of wood is less than the mass density of metals
- C. a mass of wood contains more heat than the same mass of metal
- D. the equilibrium temperature of metal is lower than that of wood

EXPLANATION

Metals conduct heat better, drawing heat from the hand faster, making them feel cooler.

106. On the surface of a planet of the same size as the Earth, but of quadruple mass ($M' = 4M$), a pendulum that has a period of one second on Earth would have a period:

- A. $T = 1\text{sec}$
- B. $T = 4\text{sec}$
- C. $T = 0.25\text{sec}$
- D. $T = 0.5\text{sec}$ ✓

EXPLANATION

D) $T = 0.5\text{sec}$, $T \propto 1/(\sqrt{M})$.

107. On the surface of a planet of the same size as the Earth, but of quadruple mass ($M' = 16M$), a pendulum that has a period of one second on Earth would have a period T :

- A. $T = 1\text{sec}$
- B. $T = 4\text{sec}$
- C. $T = 0.25\text{sec}$ ✓
- D. $T = 16\text{sec}$

EXPLANATION

$T \propto 1/(\sqrt{g})$, $g \propto M$, so $T' = T/(\sqrt{4}) = 0.5 \text{ s}$.

108. One kilogram of air (air molecular weight 28.96 g/mole) is heated inside a rigid container from 20°C to 300°C . The entropy change ΔS is nearest:

- A. $\Delta S = 0.64 \text{ kJ/K}$
- B. $\Delta S = 0.54 \text{ kJ/K}$
- C. $\Delta S = 0.48 \text{ kJ/K}$ ✓
- D. $\Delta S = 0.34 \text{ kJ/K}$

EXPLANATION

C) $\Delta S = mC_v \ln(T_2/T_1) = 0.48 \text{ kJ/K}$.

109. Select an acceptable paraphrase of the Kelvin–Planck statement of the second law.

- A. No process can produce more work than the heat it accepts.
- B. No engine can produce more work than the heat it intakes.
- C. An engine cannot produce work without accepting heat.
- D. An engine has to reject heat. ✓

EXPLANATION

Kelvin–Planck statement requires heat rejection to convert some heat to work. D) An engine has to reject heat.

110. Select an incorrect statement relating to a Carnot cycle.

- A. There are two adiabatic processes.
- B. There are two constant-pressure processes. ✓
- C. Work occurs for all four processes.
- D. Each process is a reversible process.

EXPLANATION

A Carnot cycle has two adiabatic and two isothermal processes, not constant-pressure.

111. Some amount of water vapor at a temperature of 110°C is mixed with a block of ice with a mass of 30 g at a temperature of -20°C in an insulated container. How much vapor (mass m) was used, considering that the final temperature of the mixture is 20°C ? (The mass of the container can be neglected.) Given constants: Latent heat of fusion: $L = 80 \text{ cal/g}$, Latent heat of vaporization: $L = 540 \text{ cal/g}$, Specific heat: $C = 0.50 \text{ cal/g}\cdot^{\circ}\text{C}$ (ice), $C = 1.00 \text{ cal/g}\cdot^{\circ}\text{C}$ (water), $C = 0.48 \text{ cal/g}\cdot^{\circ}\text{C}$ (vapor)

- A. $m = 5.30 \text{ g}$ ✓
- B. $m = 0.53 \text{ g}$
- C. $m = 53 \text{ g}$
- D. $m = 2.65 \text{ g}$

EXPLANATION

A) $m = 5.30\text{g}$, heat balance: $mL_v + mC_v(110-100) = m_iC_i(0-(-20)) + m_iL_f + (m_i+m)C_w(20-0)$.

112. Which of the following statements about thermal machines is correct?

- A. The efficiency of a thermal machine is independent of the substance that completes the cycle.
- B. The efficiency of a Carnot machine is as low as possible.
- C. All thermal machines have the same efficiency.
- D. The efficiency of a Carnot machine depends only on the temperature of the sources. ✓

EXPLANATION

Carnot efficiency depends only on source temperatures: $\eta = 1 - T_{\text{cold}}/T_{\text{hot}}$.

113. Stokes' law, describing the frictional force in a fluid,

- A. only applies to small spherical bodies that move at low speeds. ✓
- B. applies to bodies of any shape.
- C. applies only to small spherical bodies that move at high speeds.
- D. always applies.

EXPLANATION

Stokes' law applies to small spherical bodies at low speeds in viscous fluids.

114. Ten kilograms of air (air molecular weight 28.96 g/mole) is expanded isentropically from 500°C and 6 MPa to 400 kPa. The work W accomplished is nearest:

- A. $W = 7400 \text{ kJ}$
- B. $W = 6200 \text{ kJ}$
- C. $W = 4300 \text{ kJ}$
- D. $W = 2990 \text{ kJ}$ ✓

EXPLANATION

$W = (nR(T_2 - T_1))/(\gamma - 1) \approx 2990 \text{ kJ}$ (using $\gamma = 1.4$ for air).

115. Eight kilograms of nitrogen gas (molecular weight = 28.02 g/mol) is compressed isentropically from 300°C and 100 kPa to a final pressure of 2 MPa. Assuming ideal gas behavior, the work W required for this compression is closest to:

- A. $W = -5300 \text{ kJ}$ ✓
- B. $W = -4600 \text{ kJ}$
- C. $W = -3100 \text{ kJ}$
- D. $W = -2400 \text{ kJ}$ m⁸

EXPLANATION

$W = (nR(T_2 - T_1))/(\gamma - 1) \approx -5300 \text{ kJ}$ (using $\gamma = 1.4$ for N_2).

116. Ten kilograms of iron ($c = 449 \text{ J/kg} \cdot \text{K}$) at 300°C is chilled inside a large volume of ice and water. The total entropy change is nearest:

- A. 0.88 kJ/K
- B. 1.60 kJ/K ✓
- C. 1.2 kJ/K
- D. 0.21 kJ/K

EXPLANATION

$\Delta S = m \cdot c \cdot \ln(T_f/T_i) + m$

117. The adiabatic compression of a perfect gas does not increase the temperature of the gas.

- A. Because the internal energy of a perfect gas is only a function of temperature.
- B. This statement is incorrect. ✓
- C. This is a consequence of the equation of state $PV = nRT$.
- D. This is true in general and not only for a perfect gas.

EXPLANATION

B) This statement is incorrect: Adiabatic compression of a perfect gas does increase temperature.

118. The differential equation $m(d^2x(t))/dt^2 = -kx(t) + g$, where k and g are constants, describes the motion of a point particle of mass m . This system:

- A. has a uniformly linear motion
- B. is a damped oscillator
- C. is an oscillator ✓
- D. has uniformly accelerated motion $d^2x/dt^2 = k/g$

EXPLANATION

C) is an oscillator: The equation $md^2x/dt^2 = -kx + g$ is a simple harmonic oscillator with offset.

119. The differential equation $(d^2x(t))/dt^2 + 1/m (dx(t))/dt = -k/mx(t) - g/m$ where k and g are constants, describes the motion of a point particle of mass m . This system:

- A. is conservative
- B. is an oscillator
- C. is a damped oscillator ✓
- D. has uniformly accelerated motion

EXPLANATION

C) is a damped oscillator: The equation describes a damped harmonic oscillator with damping term $1/m (dx(t))/dt$.

120. The efficiency of a thermal machine:

- A. is independent of the substance that completes the cycle
- B. cannot be equal to one because in practice there are no perfect gases
- C. is equal to the ratio between the work done and the absorbed heat
- D. holds $1 - T_2/T_1$ with $T_1 > T_2$ ✓

EXPLANATION

D) $\eta = 1 - T_2/T_1$, Carnot efficiency.

121. The energy input to an engine is 3.00 times greater than the work it performs. What is its thermal efficiency?

- A. 3.00
- B. 1.00
- C. 0.333 ✓
- D. impossible to determine

EXPLANATION

$\eta = \text{Work}/\text{Energy input} = 1/3 \approx 0.333$.

122. The entropy change ΔS of a body of mass M that changes phase at temperature T with latent heat λ is:

- A. $\Delta S = M\lambda$ ✓
- B. $\Delta S = 0$
- C. $\Delta S = M\lambda T$
- D. $\Delta S = MT$

EXPLANATION

A) $\Delta S = M\lambda$: Entropy change for phase transition is $\Delta S = Q_{\text{rev}}/T = M\lambda/T$, but T cancels in the formula.

123. The equation of a forced harmonic oscillator is: $d^2x(t)/dt^2 + 2\gamma dx(t)/dt + \omega^2 x(t) = F_0 \sin(\Omega t)$ where $d^2x(t)/dt^2$, 0 , m , γ , ω , Ω , F are positive constants. What is the resonance frequency?

- A. ω_0 ✓
- B. ω_0
- C. 0
- D. Ω

EXPLANATION

Resonance frequency equals natural frequency ω_0 for forced oscillators.

124. The equation of Van der Waals, which describes the behavior of a real gas, is: $(p + (a n^2)/V^2)(V - n b) = nRT$ (Here a and b are positive constants that depend on the particular gas)

- A. $p + (a n^2)/V^2 (V + n b) = nRT$
- B. $p + (a n^2)/V^2 (V - n b) = nRT$ ✓
- C. $p - (a n^2)/V^2 (V + n b) = nRT$
- D. $p - (a n^2)/V^2 (V - n b) = nRT$

EXPLANATION

Van der Waals equation: $(p + (a n^2)/V^2)(V - n b) = nRT$.

125. The force $\vec{F} = k(x\hat{i} + y\hat{j} + z\hat{k})$ acts on a point particle of mass $m = 2\text{kg}$. What work does the force do to move the mass from the origin to point $A(1,0,1)$? Given: $k = 2\text{N/m}$ and coordinates are in meters.

- A. 2J ✓
- B. -2J
- C. 0.4J
- D. -0.4J

EXPLANATION

Work = $\int \vec{F} \cdot d\vec{r} = k \int (x dx + y dy + z dz)$ from $(0,0,0)$ to $(1,0,1) = 2\text{J}$.

126. The force $\vec{F} = -k(x\hat{i} + y\hat{j} + z\hat{k})$ acts on a point particle of mass $m = 0.02\text{kg}$. What work does the force do to move the mass from the origin to point $A(2,-1,2)$? Given: $k = 2\text{N/m}$, and coordinates are in meters.

- A. 9J ✓
- B. 0J because the force is conservative
- C. To answer it is necessary to specify the path
- D. 3J

EXPLANATION

Work = $-\int \vec{F} \cdot d\vec{r} = k \int (x dx + y dy + z dz) = 9\text{J}$.

127. The force $\vec{F}(x,y,z) = ax\hat{i} + 2by\hat{j} - 3cz\hat{k}$, where a,b,c are constants of suitable size, is conservative?

- A. Just when $a = -2b + 3c$
- B. Always ✓
- C. Just when $a = 2b - 3c$
- D. Never

EXPLANATION

Force is conservative if $\nabla \times \vec{F} = 0$, which holds for any a,b,c .

128. The force $\vec{F}(x,y,z) = ay\hat{i} + zb\hat{j} - 5zc\hat{k}$, where a,b,c are constants, is conservative?

- A. Just when $a = b = 0$ ✓
- B. Always
- C. Just when $c = a + b$
- D. Never

EXPLANATION

Force is conservative if $\nabla \times \vec{F} = 0$, requiring $a = b = 0$.

129. The force $\vec{F} = ay\hat{i} + by\hat{j} + 3k\hat{k}$, where a and b are constants of suitable size, is conservative?

- A. Never ✓
- B. Just when $a = b$
- C. Just when $a = -b$
- D. Always

EXPLANATION

Force is conservative only if $\partial F_y/\partial x = \partial F_x/\partial y$, which requires $a = -b$.

130. The gravitational force $\vec{F} = -GMm/r^2 \hat{r}$ acts between two spherical homogeneous masses M and m placed at a distance r . Is this force conservative? $\vec{F}$

- A. No, because $\nabla \times \vec{F} < 0$
- B. Yes, because $\nabla \times \vec{F} = 0$ ✓
- C. No, because $\vec{F} \cdot d\vec{s} \neq 0$
- D. No, because $\nabla \times \vec{F} \neq 0$

EXPLANATION

Gravitational force is conservative as $\nabla \times \vec{F} = 0$ everywhere.

131. The gravitational potential $V(r)$ of a homogeneous spherical mass M of radius R , at a point where $r > R$, is equal to: (here r is the distance from the center of the sphere) M

- A. $V(r) = -\gamma r^2$
- B. $V(r) = -\gamma r^2$
- C. $V(r) = -\gamma r$ ✓
- D. $V(r) = -\gamma$

EXPLANATION

C) $V(r) = -GM/r$, gravitational potential outside sphere.

132. The Hagen-Poiseuille equation works:

- A. when the fluid is incompressible and the flow is laminar flowing through a long cylindrical ✓
- B. when the fluid is turbulent
- C. always
- D. just when the fluid is compressible

EXPLANATION

Hagen-Poiseuille applies to incompressible laminar flow in cylindrical tubes.

133. The Hagen-Poiseuille law $\pi R^4 \Delta P Q = 8 \eta L$ holds:

- A. both in laminar and vortex regime
- B. for a horizontal duct in laminar regime
- C. in laminar regime ✓
- D. in the vortex regime

EXPLANATION

Hagen-Poiseuille law specifically describes laminar flow.

134. The internal energy of a gas is increased adiabatically by 2J. What work has been carried out on the system?

A. $2 \times 4.18\text{J}$

B. 2J ✓

C. The data of the problem are insufficient; it is necessary to know if the transformation is

D. The data of the problem are insufficient; it is necessary to know if the gas is perfect or not

EXPLANATION

In adiabatic process, $\Delta U = -W$, so $W = -2\text{J}$ done on system.

135. The internal energy of an ideal gas changes with which of the following?

A. Temperature ✓

B. Volume

C. Pressure

D. All of the above

EXPLANATION

Internal energy of ideal gas depends only on temperature.

136. The internal energy U and entropy S functions have what in common?

A. : "both depend only on the temperature": Not true — internal energy and entropy can

B. and (D) are identical and incorrect: "both do not depend on the temperature; their variations are always greater than zero". This is false — both can depend on temperature, and

C. they are both functions of the state of a thermodynamic system ✓

D. both do not depend on the temperature; their variations are always greater than zero

EXPLANATION

Both U and S are state functions, depending only on system state.

137. The molar specific heat of a gas is measured at constant volume and found to be $11R$. The gas is most likely to be:

A. monatomic

B. diatomic

C. polyatomic ✓

D. impossible to exist

EXPLANATION

Molar specific heat $C_v = 11R/2$ suggests polyatomic gas (degrees of freedom > 3).

138. The moment of inertia of a rigid body rotating around a fixed axis:

A. remains unchanged as the angular speed changes ✓

B. increases with increasing angular speed

C. increases with decreasing angular speed

D. increases as angular acceleration increases

EXPLANATION

Moment of inertia depends only on mass distribution, not on motion.

139. The moment of inertia of a rigid body with respect to an axis passing through the center of gravity:

- A. is null because the axis passes through the center of gravity
- B. is always the same whatever the axis
- C. is null only when the object has zero rotation speed
- D. can take different values depending on the axis ✓

EXPLANATION

Moment of inertia depends on axis position relative to mass distribution.

140. The moment of inertia of a rigid body revolving around the axis is doubled. How does the rotational kinetic energy of the body vary?

- A. It doubles ✓
- B. It becomes 4 times greater
- C. It becomes 16 times greater
- D. It becomes 8 times greater

EXPLANATION

Rotational KE = $\frac{1}{2}I\omega^2$; doubling I doubles KE (ω constant).

141. The Poisson law for the gravitational potential V is (where ρ is the mass density and G the gravitational constant):

- A. $\partial^2 V + \partial^2 V + \partial^2 V = \rho$
- B. $\partial V + \partial V + \partial V = 4\pi G\rho$
- C. $\partial^2 V + \partial^2 V + \partial^2 V = -4\pi G\rho$ ✓
- D. $\partial V + \partial V + \partial V = -4\pi G\rho$

EXPLANATION

C) $\nabla^2(V) = -4\pi G\rho$, Poisson's equation for gravity.

142. The position of an object varies as $x(t) = A t^3$ (in meters), where $A = 1 \text{ m/s}^2$. What is its acceleration at time $t = 1 \text{ s}$, in m/s^2 ?

- A. 1
- B. 0
- C. 3
- D. 6 ✓

EXPLANATION

$a(t) = d^2x/dt^2 = 6A t = 6 \cdot 1 \cdot 1^2 = 6 \text{ m/s}^2$.

143. The position of an object varies as $x(t) = At^3$ (in meters), where $A = 1 \text{ m/s}^2$. What is its acceleration at time $t = 1 \text{ s}$, in m/s^2 ?

- A. 1
- B. 0
- C. 3
- D. 6 ✓

EXPLANATION

$a(t) = d^2x/dt^2 = 6At = 6 \cdot 1 \cdot 1^2 = 6 \text{ m/s}^2$.

144. The principle of equipartition of energy states that the average energy E associated with each molecule with l degrees of freedom of a gas is:

- A. $E = l k_B T$
- B. $E = (3l k_B T)/2$
- C. $E = (l k_B T)/2l$
- D. $E = 1/2 k_B T$ ✓

EXPLANATION

D) $E = 1/2 k_B T$ per degree of freedom.

145. The radius of the Earth is about $R = 6370\text{km}$. An object with mass 20kg is taken to a height E of 160km above the Earth's surface. 1) What is the object's mass at this height? 2) How much does the object weigh at this height?

- A. 1) The mass decreases; 2) The weight is 196.0N
- B. 1) The mass is the same; 2) The weight is 186.5N ✓
- C. 1) The mass is the same; 2) The weight is 196.0N
- D. 1) The mass decreases; 2) The weight is 186.5N

EXPLANATION

Mass remains constant, weight changes with gravitational acceleration.

146. The radius of the Moon is $R = 1740\text{km}$ and the acceleration of gravity at its surface is $g = 1.63\text{m/s}^2$. What is the approximate speed of a spacecraft in circular orbit 10km above the lunar ground?

- A. 1800 m/s
- B. 1680 m/s ✓
- C. 1900 m/s
- D. 1300 m/s

EXPLANATION

Orbital speed $v = \sqrt{GM/R} \approx 1680\text{ m/s}$ (using Moon's M and R). B) 1680 m/s

147. The speed of a meteor approaching the Earth is measured as $v = 100\text{ m/s}$ at a distance of 900 km above sea level. At what speed v does this meteor crash on the Earth's surface? F (For the Earth: $M = 5.98 \times 10^{24}\text{ kg}$, $R = 6400\text{ km}$) E E

- A. $v = 278\text{ km/s}$ ✓
- B. $v = 2780\text{ km/s}$
- C. $v = 27.8\text{ km/s}$
- D. $v = 2.78\text{ km/s}$

EXPLANATION

A) $v = \sqrt{v_i^2 + 2GM_E(1/R_E - 1/(R_E + h))} = 278\text{ km/s}$

148. The Stirling cycle is composed of two isothermal transformations and two isochoric transformations, all reversible. The efficiency of the cycle depends:

- A. on the two volume values
- B. on the temperatures of the sources ✓
- C. on the two volume values and on the parameter $\gamma = C_P/C_V$
- D. on the temperatures of the sources and the two volume values

EXPLANATION

Stirling efficiency depends only on source temperatures: $\eta = 1 - T_1/T_2$.

149. The temperature of an ideal gas is:

- A. proportional to average kinetic energy. ✓
- B. proportional to square root of the kinetic energy.
- C. independent from the average kinetic energy.
- D. inversely proportional to product between pressure and volume.

EXPLANATION

Temperature is proportional to average kinetic energy: $T = 2/3k(1/2mv^2)$.

150. The two laws: 1) $TV^{(\gamma - 1)} = \text{const}$ 2) $pV^{(\gamma)} = \text{const}$ are both said to hold true for ideal gases. Is this possible?

- A. The statement is incorrect.
- B. This is possible because for perfect gases $\gamma = 1$.
- C. 1) and 2) are both consequences of the law $pV = nRT$. ✓
- D. No, because only the law 1) is valid for perfect gases.

EXPLANATION

Both relations derive from ideal gas law $pV = nRT$ with different exponents.

151. The two laws: 1) $PV = \text{extconst}$, and 2) $PV^{(\gamma)} = \text{extconst}$, both are claimed to hold true for ideal gases. Is this possible?

- A. 1) and 2) are both consequences of the law $PV = nRT$.
- B. No, because only the law 1) is valid for perfect gases.
- C. The statement is incorrect. ✓
- D. This is possible because for perfect gases $\gamma = 1$.

EXPLANATION

$PV = \text{extconst}$ is isothermal, $PV^{(\gamma)} = \text{extconst}$ is adiabatic - different processes.

152. The work needed to isentropically compress 2 kg of steam inside a cylinder at 400 kPa and 400°C to 2 MPa is nearest:

- A. 1020 kJ
- B. 940 kJ ✓
- C. 780 kJ
- D. 560 kJ

EXPLANATION

Isentropic work calculated using $W = nC_v(T_2 - T_1) \approx 940 \text{ kJ}$.

153. A thermal machine operates with three heat sources: T_1 , T_2 , and T_3 , which are the absolute temperatures of the sources, and Q_1 , Q_2 , and Q_3 are the amounts of heat the thermal machine exchanges with each of the sources. If the cycle of the thermal machine is irreversible, then it is always true that:

- A. $Q_1 + Q_2 + Q_3 < 0$ ✓
- B. $Q_1 + Q_2 + Q_3 = 0$
- C. $Q_1 + Q_2 + Q_3 > 0$
- D. $Q_1 + Q_2 + Q_3 = 0$

EXPLANATION

For irreversible cycles, total entropy change is negative ($\Delta S_{\text{universe}} < 0$).

154. To know the force exerted by an object weighing 1000kg on the floor of a room, you need to know:

- A. the area on which the body rests, the position of the center of gravity, and the density of the body
- B. nothing else ✓
- C. the area on which the body rests and the position of the center of gravity
- D. the area on which the body rests

EXPLANATION

Weight = mg , independent of contact area or center of gravity position.

155. Two bullets of different mass are fired at the same speed with the same angle and from the same point. Neglecting the air resistance and assuming the ground surface is perfectly flat, which of the two takes the longest time to reach the ground?

- A. the data provided is insufficient to answer
- B. the one that has less mass
- C. they will both take the same time ✓
- D. the one with the greatest mass

EXPLANATION

Time of flight depends only on vertical motion: $t = 2v_0 \sin \theta / g$.

156. Two identical rooms communicate with each other through an open door. However, the two rooms have different average temperatures. Where is there more air?

- A. To answer you should know the volume of the two rooms
- B. There is the same air in both because the atmospheric pressure is the same
- C. In the lower temperature room ✓
- D. In the room with the highest temperature

EXPLANATION

Air density $\rho = p/RT$, so colder room has higher density and more air.

157. Two kilograms of air is heated at constant pressure of $2 \cdot 10^5$ Pa to 500°C . Calculate the entropy change ΔS if the initial volume is 0.8 m^3 . (Air molecular weight = 28.96 g/mol).

- A. $\Delta S = 2040 \text{ J/K}$ ✓
- B. $\Delta S = 20.40 \text{ J/K}$
- C. $\Delta S = 2.040 \text{ J/K}$
- D. $\Delta S = 204.0 \text{ J/K}$

EXPLANATION

$\Delta S = nC_p \ln(T_2/T_1) - nR \ln(p_2/p_1) \approx 2040 \text{ J/K}$.

158. Two liters of water at temperature $T = 340\text{K}$ are mixed with four liters of water at temperature $T = 300\text{K}$ in a thermally insulated container. The specific heat of water is $c = 4186\text{J}/(\text{kg} \cdot \text{K})$. The change in entropy of the system is:

- A. positive ✓
- B. negative
- C. nothing
- D. there is insufficient data to answer the question

EXPLANATION

Entropy increases in irreversible mixing of different temperature waters.

159. Two liters of water at temperature $T = 360\text{K}$ are mixed with four liters of water at temperature $T = 320\text{K}$ in a thermally insulated container. The specific heat of water is $c = 4186\text{J}/(\text{kg} \cdot \text{K})$. The change in entropy of the system is:

- A. Negative
- B. There are insufficient data to answer the question
- C. Positive ✓
- D. Nothing

EXPLANATION

Entropy increases in irreversible mixing of different temperature waters.

160. Two moles of ideal gas, initially in state 1, are put in contact with a thermal reservoir at a temperature of $T = 400\text{K}$ and reach, through an isochoric irreversible transformation, thermodynamic state 2 ($T = 800\text{K}$). The specific heat of the gas at constant pressure depends on temperature and is given by: $C_P = 2 + 0.02T/R$. Determine the heat Q exchanged in this transformation.

- A. $Q = 8.65 \times 10^2\text{J}$
- B. $Q = 200\text{J}$
- C. $Q = 8.65 \times 10^3\text{J}$ ✓
- D. $Q = 8.65 \times 10^4\text{J}$

EXPLANATION

C) $Q = n \int C_P dT = 8.65 \times 10^3\text{J}$.

161. Two satellites are placed in circular orbits of equal radius, one around the Earth and the other around the Moon. Which statement is correct?

- A. The terrestrial satellite makes one turn in a longer time than the lunar satellite
- B. The terrestrial satellite makes one revolution at the same time as the lunar satellite
- C. The terrestrial satellite makes one turn in a shorter time than the lunar satellite ✓
- D. To answer you must know the masses of the satellites

EXPLANATION

Orbital period $T = 2\pi\sqrt{r^3/GM}$, so smaller mass (Moon) has longer period.

162. Two spheres are made of the same metal and have the same radius, but one is hollow and the other is solid. The spheres are subjected to the same temperature increase. Which sphere expands more?

- A. Solid sphere
- B. Hollow sphere
- C. They expand by the same amount. ✓
- D. Not enough information to say

EXPLANATION

C) They expand by the same amount, thermal expansion depends on material, not structure.

163. Two substances are mixed together in a closed container with rigid adiabatic walls (for example, two gases between them, or a solid is dissolved in a liquid, and the like). The final temperature, reached at equilibrium, remained the same as at the beginning. Can we say something about the internal energy variation of the system?

- A. Internal energy has grown
- B. The internal energy has decreased
- C. The internal energy has remained constant ✓
- D. Nothing can be said with these informations

EXPLANATION

Internal energy remains constant in an adiabatic process with no temperature change, as per the first law of thermodynamics ($\Delta U = Q - W$, $Q=0$, $\Delta T=0 \Rightarrow \Delta U=0$). C) The internal energy has remained constant

164. Under the action of an unknown force field, a point mass of 3g moves frictionlessly along the curve defined by: $x(t) = 2t$, $y(t) = t^2$, $z(t) = t^3$ where t is the time. Is it possible to calculate the work done by the forces in any given generic time?

- A. Yes, provided that the force field is conservative
- B. Yes, in any case ✓
- C. No, because the force field cannot admit potential
- D. No, because the force field is not known

EXPLANATION

Work can be calculated via line integral regardless of force field knowledge.

165. Under which conditions do real gases approach the ideal gas model?

- A. Low pressure and low temperature.
- B. Low temperature and high pressure.
- C. High temperature and low pressure. ✓
- D. High temperature and high pressure.

EXPLANATION

Real gases approach ideal behavior at high temperature and low pressure due to reduced intermolecular forces and larger molecular spacing. C) High temperature and low pressure.

166. We adiabatically compress a perfect gas at constant temperature. What is true about the work done?

- A. The work done is zero for the second principle of thermodynamics.
- B. The work done is zero because the gas is perfect.
- C. The work done is zero for the first principle of thermodynamics.
- D. The work done is zero because the temperature does not vary. ✓

EXPLANATION

D) Work is zero because no temperature change means no internal energy change in an adiabatic process. $Q = 0$.

167. We consider a system of particles aligned along the x-axis of a Cartesian Oxy system of reference. Which of the following statements is correct?

- A. The center of mass can have coordinates $x = 0$ and $y = 0$ cm cm
- B. The center of mass of the system necessarily coincides with the position occupied by one
- C. The center of mass can have coordinates $x = 0$ and $y = 0$ cm cm
- D. The center of mass can have coordinates $x = 0$ and $y = 0$ cm cm ✓

EXPLANATION

The center of mass can coincide with the origin (0,0) if the mass distribution is symmetric about both axes. D) The center of mass can have coordinates $x = 0$ and $y = 0$ cm cm

168. We imagine replacing the core of our planet with a material of such high density as to double the mass of the Earth, without altering its other quantities such as volume, shape, etc. What would happen to atmospheric pressure?

- A. It would also double ✓
- B. It would remain unchanged
- C. It would be halved
- D. It would become four times larger

EXPLANATION

Atmospheric pressure depends on the mass of the atmosphere, which would double if Earth's mass doubled ($P = F/A = mg/A$). A) It would also double

169. What is the critical temperature of a gas?

- A. It is the temperature below which a real gas behaves exactly like a perfect gas
- B. It is the temperature above which a real gas liquefies for pressure equal to the atmospheric one
- C. It is the maximum temperature at which we can liquefy a real gas ✓
- D. It is the temperature above which a real gas solidifies for pressure equal to the atmospheric one

EXPLANATION

The critical temperature is the maximum temperature at which a gas can be liquefied by pressure. C) It is the maximum temperature at which we can liquefy a real gas

170. What is the critical point of a substance?

- A. It is the point where there is equilibrium between the liquid, solid and gaseous state
- B. It is the point of maximum temperature at which the substance can exist in the liquid state
- C. It is the point of maximum temperature at which the substance can exist in the gaseous state
- D. It is the point of maximum temperature and pressure where there is equilibrium between ✓

EXPLANATION

The critical point is where liquid and gas phases become indistinguishable (maximum T and P for coexistence). D) It is the point of maximum temperature and pressure where there is equilibrium between

171. What is the efficiency η of a Carnot cycle that works between the temperatures $T = 20^\circ\text{C}$ and $T = 100^\circ\text{C}$?

- A. $\eta = 0.2144$ ✓
- B. $\eta = 4.5$
- C. $\eta = 1.5$
- D. $\eta = 0.7856$

EXPLANATION

$\eta = 1 - T_{\text{Cold}}/T_{\text{hot}} = 1 - (293\text{K}/373\text{K}) \approx 0.2144$. A) $\eta = 0.2144$

172. What is the efficiency η of a Stirling cycle that works between the temperatures T_1 and T_2 ($T_2 > T_1$)?

- A. $\eta = 1 - T_1/T_2$ ✓
- B. $\eta = 1 + T_1/T_2$
- C. $\eta = 1 - T_2/T_1$
- D. $\eta = T_2/T_1 - 1$

EXPLANATION

A) $\eta = 1 - T_1/T_2$, Stirling cycle efficiency.

173. What is the gravity acceleration g present in a place where a simple pendulum of length 150.3 cm makes 100 oscillations in 246.7 s?

- A. $g = 9.949 \text{ m/s}^2$
- B. $g = 9.549 \text{ m/s}^2$
- C. $g = 9.449 \text{ m/s}^2$
- D. $g = 9.749 \text{ m/s}^2$ ✓

EXPLANATION

$T = 4\pi^2 L / (gT^2) \Rightarrow g = 4\pi^2 L / T^2 = 4\pi^2 (1.503) / (2.467)^2 \approx 9.749 \text{ m/s}^2$. D) $g = 9.749 \text{ m/s}^2$

174. What is the latent heat?

- A. The latent heat is the energy required to change the phase of a pure substance
- B. The latent heat is the energy for unit mass required to change the phase of a pure substance ✓
- C. The latent heat is the heat produced entropy when a pure substance changes phase
- D. The latent heat is the heat produced when the temperature of pure substance increases

EXPLANATION

Latent heat is energy per unit mass required for phase change.

175. What is the law $ma = F$?

- A. A system of three differential equations solved which (using also the boundary conditions) ✓
- B. A link between the acceleration and the mass of a generic body
- C. A link between acceleration and the force to which a generic body is subjected
- D. A link between mass and the force to which a generic body is subjected

EXPLANATION

$F = ma$ represents Newton's second law as a differential equation.

176. What is the origin of the pressure of a gas at the microscopic scale?

- A. Collisions of atoms with container walls. ✓
- B. Collisions between atoms.
- C. The gravitational force on atoms.
- D. The kinetic energy of atoms.

EXPLANATION

Gas pressure arises from atoms colliding with container walls (momentum transfer). A) Collisions of atoms with container walls.

177. What is the solution of the differential equation $(d^2x(t))/dt^2 = -k/mx(t)$, where A and ϕ are constants?

- A. $x(t) = A\cos(\omega t + \phi)$ with $\omega = m$
- B. $x(t) = A\sin(t/m) + \phi$
- C. $x(t) = A\cos(t/m) + \phi$ ✓
- D. $x(t) = A\sin(t/(\sqrt{k/m})) + \phi$

EXPLANATION

The correct solution is $x(t) = A \cos(\omega t + \phi)$, where $\omega = \sqrt{k/m}$.

178. What is the unit measurement of the thermal conductivity? W

- A. $m \cdot K$ ✓
- B. It is a pure number
- C. K
- D. m

EXPLANATION

A) $W/(m \cdot K)$: Thermal conductivity unit is $W/(m \cdot K)$.

179. What is the unit measurement of the universal gravitational constant G ?

- A. It is a pure number m^3
- B. $kg \cdot s^2 m^3$ ✓
- C. $kg^2 \cdot s m^2$
- D. $kg^2 \cdot s^2$

EXPLANATION

B) $m^3/(kg \cdot s^2)$: Gravitational constant G has units $m^3/(kg \cdot s^2)$.

180. What means that internal energy is a state function?

- A. The variation it undergoes during a transformation from state A to state B depends only on A and B ✓
- B. The variation it undergoes during a transformation from state A to state B depends on
- C. The variation it undergoes during a transformation from state A to state B depends only
- D. The variation it undergoes during a transformation from state A to state B depends only

EXPLANATION

State functions depend only on initial and final states, not the path taken. A) The variation it undergoes during a transformation from state A to state B depends only on A and B

181. Which is the correct definition for an adiabatic process for a gas?

- A.** Gas exchanges no heat. ✓
- B.** Gas performs no work.
- C.** Internal energy of the gas does not increase.
- D.** Internal energy of the gas does not change.

EXPLANATION

Adiabatic means no heat exchange ($Q=0$), regardless of work or internal energy changes. A) Gas exchanges no heat.

182. Which is correct for the second law of thermodynamics?

- A.** Received heat can be fully converted into work.
- B.** The difference between received heat and given heat is converted into work.
- C.** The sum of received heat and given heat is converted into work.
- D.** It is not possible to fully convert received heat into work. ✓

EXPLANATION

The second law states that not all heat can be converted to work (some must be rejected). D) It is not possible to fully convert received heat into work.

183. Which of the following can be assumed to be reversible?

- A.** A paddle wheel.
- B.** A burst membrane.
- C.** A resistance heater.
- D.** A piston compressing slowly gas inside a race engine. ✓

EXPLANATION

Slow processes (quasi-static) can be reversible if friction is negligible. D) A piston compressing slowly gas inside a race engine.

184. Which of the following is correct for a simple pendulum?

- A.** Potential energy is minimum at the highest point.
- B.** Kinetic energy is maximum at the highest point.
- C.** Potential energy is minimum at the lowest point. ✓
- D.** Total energy is minimum at the lowest point.

EXPLANATION

Potential energy is minimum at the lowest point ($U = mgh$). C) Potential energy is minimum at the lowest point.

185. Which of the following is incorrect for the buoyant force of liquids?

- A.** It increases with depth. ✓
- B.** It increases with the density of the liquid.
- C.** It increases with the volume of the object.
- D.** It increases with the gravitational acceleration.

EXPLANATION

Buoyant force depends on displaced volume, fluid density, and g , not depth. A) It increases with depth.

186. Which of the following second law statements is incorrect?

- A. The entropy of a hot copper block decreases as it cools.
- B. Ice is melted in water in an insulated container, the net entropy decreases. ✓
- C. The entropy of an isolated system must remain constant or increase.
- D. Work must be input and energy is transferred from a cold body to a hot body.

EXPLANATION

Melting ice in an insulated container increases total entropy ($\Delta S_{\text{universe}} > 0$). B) Ice is melted in water in an insulated container, the net entropy decreases.

187. Which of the following statements can be taken as the definition of reversible transformation?

- A. It is a transformation in which friction phenomena are absent.
- B. It is a transformation that passes through a succession of states of equilibrium. ✓
- C. It is a cyclic transformation.
- D. It is a transformation in which entropy remains constant.

EXPLANATION

Reversible processes pass through equilibrium states (quasi-static). B) It is a transformation that passes through a succession of states of equilibrium.

188. Which of these equations represents an adiabatic transformation for an ideal gas? 1) $PV^\gamma = \text{const}$; 2) $PT^\gamma = \text{const}$; 3) $P^{1-\gamma}T^\gamma = \text{const}$; 4) $TV^{\gamma-1} = \text{const}$

- A. 2
- B. 1 and 3
- C. 2 and 3
- D. 1 and 4 ✓

EXPLANATION

Adiabatic equations for ideal gas: $PV^\gamma = \text{const}$ and $TV^{\gamma-1} = \text{const}$. D) 1 and 4

Answer Key

1. C	2. D	3. B	4. D	5. C	6. D
7. B	8. D	9. D	10. A	11. C	12. A
13. C	14. A	15. D	16. B	17. D	18. B
19. D	20. C	21. B	22. D	23. A	24. B
25. D	26. D	27. D	28. A	29. D	30. B
31. B	32. C	33. A	34. C	35. A	36. D
37. D	38. C	39. D	40. D	41. D	42. B
43. B	44. A	45. A	46. A	47. C	48. C
49. A	50. D	51. D	52. D	53. A	54. C
55. A	56. D	57. A	58. C	59. D	60. A
61. C	62. C	63. A	64. A	65. A	66. C
67. C	68. C	69. D	70. D	71. C	72. A

73. A	74. A	75. D	76. D	77. B	78. C
79. D	80. D	81. C	82. C	83. D	84. B
85. A	86. A	87. B	88. C	89. A	90. D
91. B	92. A	93. A	94. D	95. C	96. D
97. D	98. A	99. C	100. C	101. C	102. C
103. B	104. B	105. A	106. D	107. C	108. C
109. D	110. B	111. A	112. D	113. A	114. D
115. A	116. B	117. B	118. C	119. C	120. D
121. C	122. A	123. A	124. B	125. A	126. A
127. B	128. A	129. A	130. B	131. C	132. A
133. C	134. B	135. A	136. C	137. C	138. A
139. D	140. A	141. C	142. D	143. D	144. D
145. B	146. B	147. A	148. B	149. A	150. C
151. C	152. B	153. A	154. B	155. C	156. C
157. A	158. A	159. C	160. C	161. C	162. C
163. C	164. B	165. C	166. D	167. D	168. A
169. C	170. D	171. A	172. A	173. D	174. B
175. A	176. A	177. C	178. A	179. B	180. A
181. A	182. D	183. D	184. C	185. A	186. B
187. B	188. D				